Interleaving recovers the full circuit distance of the gross and $[[90, 8, 10]]$ bivariate-bicycle codes

Guancheng Pan[1, *]

[1] *Zhejiang University, Hangzhou, China*

(Dated: October 24, 2026)

Strikis, Browne and Beverland (arXiv:2603.05481) showed that every non-interleaved single-ancilla syndrome-extraction circuit has circuit distance at most that of an extended code: at most 11 for the gross code $[[144, 12, 12]]$ and, as we show, at most 8 for $[[90, 8, 10]]$. We construct single-ancilla circuits with a period of 11 time steps that reach the full code distance, 12 for the gross code, answering their open question, and 10 for $[[90, 8, 10]]$, two above the extended-code distance of their CNOT orders, and prove this under circuit-level depolarizing noise for every number of rounds. The method fixes an order class, even one of lower extended-code distance, and recovers the code distance by exact lift computations over complete lists of the lightest extended logicals, with every solver decision in the proofs backed by a checked proof certificate. Without idle noise the $[[90, 8, 10]]$ circuit has a lower logical error rate than IBM's at $p = 3 \times 10^{-3}$, and the gross circuit one lower than the distance-11 circuit of Strikis et al.; with standard idle noise both are worse than the IBM circuits. In a neutral-atom noise model with literature parameters the $[[90, 8, 10]]$ circuit again has the lower rate, provided the ancilla arrays move sequentially and atoms left unpaired during entangling pulses do not err.

Bivariate-bicycle (BB) codes [1] are leading candidates for low-overhead quantum memory; IBM's architecture is built on the gross code, with parameters $[[144, 12, 12]]$ [1, 2]. What a syndrome circuit delivers is its circuit distance d_{circ}, the minimum number of circuit faults that flip a logical observable without triggering any detector; at low physical error rates it governs the logical error rate.

No circuit for the gross code had been shown to reach $d_{\mathrm{circ}} = d = 12$. The IBM circuit, with 8 time steps per round, has circuit distance exactly 10 under SD6 for $R \leq 3$, by exact enumeration (SM S7); previously only $d_{\mathrm{circ}} \leq 10$ was known [1–3]. Strikis, Browne and Beverland (SBB) [4] analysed single-ancilla circuits, in which each check is measured by one dedicated ancilla coupled by CNOTs. They append to the check matrix the "residual" errors that an ancilla fault spreads onto the data, obtaining an extended code of distance d_{ext}, and show that $d_{\mathrm{circ}} \geq d_{\mathrm{ext}}$ for every such circuit (their Prop. 3), while $d_{\mathrm{circ}} \leq d_{\mathrm{ext}}$ when the circuit is *non-interleaved* in their sense, i.e. when all X-type CNOTs precede all overlapping Z-type CNOTs or vice versa (their Sec. II A and Cor. 1). Since every CNOT order gives $d_{\mathrm{ext}} \leq 11$ for the gross code, non-interleaved circuits are capped at 11 (their Prop. 4); their depth-8 circuit (their Fig. 12, hereafter "Fig. 12") was conjectured to reach it. They left open "whether any interleaved circuit can achieve circuit distance 12" [4]. For the $[[90, 8, 10]]$ code of the same family the best known value is 8: the IBM circuit has $d_{\mathrm{circ}} \leq 8$ [1], proven exact by MaxSAT [3].

We answer this question in the affirmative. For the gross code we construct a single-ancilla circuit with a period of 11 time steps that is uniformly tiled, all X checks acting identically up to translation and likewise all Z checks, and interleaved, and prove $d_{\mathrm{circ}} = 12$ for every number of rounds (Theorem 1). SBB's Prop. 5 bounds every uniformly tiled non-interleaved circuit for the gross code by $d_{\mathrm{circ}} \leq 10$ [4], so interleaving gains two units within this class, and one unit over the best non-interleaved circuit overall. For $[[90, 8, 10]]$, where every choice of CNOT orders gives $d_{\mathrm{ext}} \leq 8$ and hence every non-interleaved circuit has $d_{\mathrm{circ}} \leq 8$, we construct a period-11 circuit with $d_{\mathrm{circ}} = 10$ (Theorem 2); a period-16 circuit reaches the same distance, and the new circuit shortens the period (Table I). The method is to choose a class of CNOT orders, even one of lower d_{ext}, and to recover the code distance through timing, by deciding exactly, over the complete lists of extended logicals of weight d_{ext} and $d_{\mathrm{ext}} + 1$, that none of them can be realised by undetected circuit faults. This turns circuit design into a finite decision problem, a tool for single-ancilla circuits in general. Our computation of d_{ext} also proves SBB's conjecture that Fig. 12 has $d_{\mathrm{circ}} = 11$ (Corollary 1). The new circuits run with 11 time steps per round instead of 8; we discuss this cost at the end.

Exact circuit distances are known for hypergraph-product codes [5], for the smallest BB circuit [6], and as certificates for other qLDPC constructions [7, 8]. Shaw and Terhal's morphing circuit with the same qubit count acts on a $[[144, 12, 12]]$ code that they call "a second 'gross' code", with $d_{\mathrm{circ}} \leq 12$ [9]; a morphing idling circuit for the gross code is "believed to have circuit distance 12" [10]. We note that this end-cycle code is not equivalent to the gross code, since its minimum stabiliser weight is 9 rather than 6 (SM S19). For $[[90, 8, 10]]$ the upper bound for the unchanged IBM circuit returns to 10 under infinitely Z-biased ancilla noise [11]; we keep depolarizing noise, change the circuit, and prove the value. Using SBB's lower bound via non-interleaved counterparts to screen interleaved schedules was proposed in Ref. [12]; that bound cannot exceed d_{ext}. Staggering con-

secutive rounds is permitted in SBB's model and generalizes the pipelined product-coloration circuits of Ref. [13]; the new ingredient here is interleaving X- and Z-check CNOTs on each data qubit within a uniform, wrapped-around schedule. For the surface code, interleaving X and Z checks recovers the full distance at minimum depth [14]. Check-dependent CNOT orders are an established tool [4, 15, 16].

Setting. We consider memory experiments in the Z or X basis: data initialisation, R rounds of syndrome extraction and transversal readout. Each check has one ancilla that is reset, coupled by six CNOTs and measured once per round; rounds of different checks may overlap in time (pipelining). Faults are the single-operation Pauli components of standard circuit-level depolarizing noise (SD6 [17]). Faults with identical detector and observable effects are merged into one detector-error-model (DEM) mechanism [18], and d_{circ} counts distinct mechanisms. For a check with CNOT order $(q_1, \ldots, q_6)$, an ancilla fault after the k-th CNOT deposits the residual $\{q_{k+1}, \ldots, q_6\}$ on the data; the extended code appends all residual columns (Hs, Ls) to the data columns (H: check matrix, L: logical operators). Every single fault, propagated to the readout, has a data image that is zero or one extended-code column (SBB, Table I; proved for our template in SM S2 [19]).

The circuits. Both circuits are translation invariant: all X checks share one order and timing, as do all Z checks (Table I). Both are legal and deterministic. Within one period every data qubit performs its three X-check CNOTs and then its three Z-check CNOTs, yet neither circuit is non-interleaved: because the Z-check window wraps across the period boundary, a Z-check instance follows some X-check instances of its round and precedes others, closing cycles of precedence through both types, and no reordering of commuting gates separates the X and Z CNOTs of a round (Fig. 1a, SM S3). This is the sense in which they are interleaved, and it is what lifts d_{circ} above d_{ext}: a readout can now fall between the arrivals of two hook residuals on the same data qubit, so a check sees one of them instead of both, and an extended logical that a segmented circuit realises silently triggers a detector (Fig. 1b). All 144 weight-10 extended logicals lift in the segmented circuit and none in the period-11 circuit at $R = 1$; for each, three or four Z checks see exactly two of its hook residuals, deposited on opposite sides of the check's read (on one data qubit, as in Fig. 1b, or on two), so a single detector already forbids the lift. In the gross circuit both check types use orders of the uniform class of $A_1A_2A_3B_3B_1B_2$ (closed under reversal and the swaps of positions 1,2 and 5,6), whose extended code has $d_{\mathrm{ext}} = 10$, with exactly 144 extended logicals of weight 10 and 7344 of weight 11 per basis. In the $[[90, 8, 10]]$ circuit the orders give $d_{\mathrm{ext}} = 8$, with 585 extended logicals of weight 8 and 16830 of weight 9. Both lists were obtained by two independent exhaustive enumerations with identical results. The circuits were found by SAT searches over translation-invariant schedules with constraints that block these logicals in a bulk Pauli-frame model, followed by exact certification (SM S3, S4).

Two lemmas. The proofs rest on two lemmas valid for every circuit of our template.

Lemma 1 (Reduction) *An undetected fault set of size w that flips a logical observable yields an extended-code logical of weight at most w, formed by its nonzero fault images.*

Proof. Add one observable per row of H and L on the final data readout. The observable of row H_i equals the sum of all detectors of check i, which telescope from the first measurement to the final readout. For an undetected fault set the H parts of the images therefore sum to zero and the L parts to the logical effect; each image is zero or one column, and equal columns cancel. □

Lemma 2 (Light logicals) *An undetected logical of weight d_{ext} consists of d_{ext} faults whose images are distinct columns forming an extended logical. One of weight $d_{\mathrm{ext}} + 1$ consists either of $d_{\mathrm{ext}} + 1$ such faults, or of d_{ext} such faults forming an extended logical together with one fault of zero image.*

The proof is a two-line count (End Matter). The lift computations admit every zero-image mechanism of the DEM as a candidate; in the gross circuit these are exactly the ancilla readout flips. Deciding $d_{\mathrm{circ}} \geq d_{\mathrm{ext}} + 2$ is thus a finite problem: decide which extended logicals of weight d_{ext}, with or without one zero-image fault, and of weight $d_{\mathrm{ext}} + 1$ lift, i.e. can be realised by circuit faults with the corresponding images that trigger no detector.

Theorem 1 *Under SD6 noise, the gross-code circuit of Table I has $d_{\mathrm{circ}} = 12$ in both memory bases for every number of rounds $R \geq 1$.*

Proof sketch. By Lemma 1 with $d_{\mathrm{ext}} = 10$ no undetected logical has weight ≤ 9. By Lemma 2, one of weight 10 or 11 would lift one of the 144 weight-10 extended logicals, possibly with one readout flip, or one of the 7344 of weight 11. For $R = 1, \ldots, 31$ and both bases none of them lifts, as found by independent exact implementations (CP-SAT, a frontier dynamic program and a per-logical SAT model). A fault-free window of $G = 31$ layers would split an undetected logical into two undetected parts, one lighter, so a weight-11 logical spans at most 300 layers, and deleting fault-free periods maps it into a circuit with at most 31 rounds (End Matter). A weight-12 data logical, applied as flips after initialisation, triggers no detector and flips the observable. □ A direct branch-and-bound search on the full DEM, which uses no extended code, finds no undetected logical of weight ≤ 11 at $R = 1$ and 2 in both bases.

TABLE I. The two period-11 circuits. All checks of a type share the reset step, CNOT order, CNOT steps and measurement step (modulo 11); a Z-check window passes the period boundary and continues in the next period. Labels: $A = x^3 + y + y^2$, $B = y^3 + x + x^2$ (gross) and $A = x^9 + y + y^2$, $B = 1 + x^2 + x^7$ ($[[90, 8, 10]]$) [1], $A \mapsto A_1A_2A_3$, $B \mapsto B_1B_2B_3$.

code	type	reset	CNOT order	CNOT steps	meas.
gross $[[144, 12, 12]]$	X	0	$A_2A_1A_3B_3B_2B_1$	1, 2, 3, 4, 5, 9	10
	Z	10	$B_1B_2B_3A_3A_1A_2$	0, 4, 5, 6, 7, 8	9
$[[90, 8, 10]]$	X	0	$B_1B_3B_2A_3A_1A_2$	1, 2, 3, 4, 5, 9	10
	Z	10	$A_2A_1A_3B_2B_3B_1$	0, 4, 5, 6, 7, 8	9

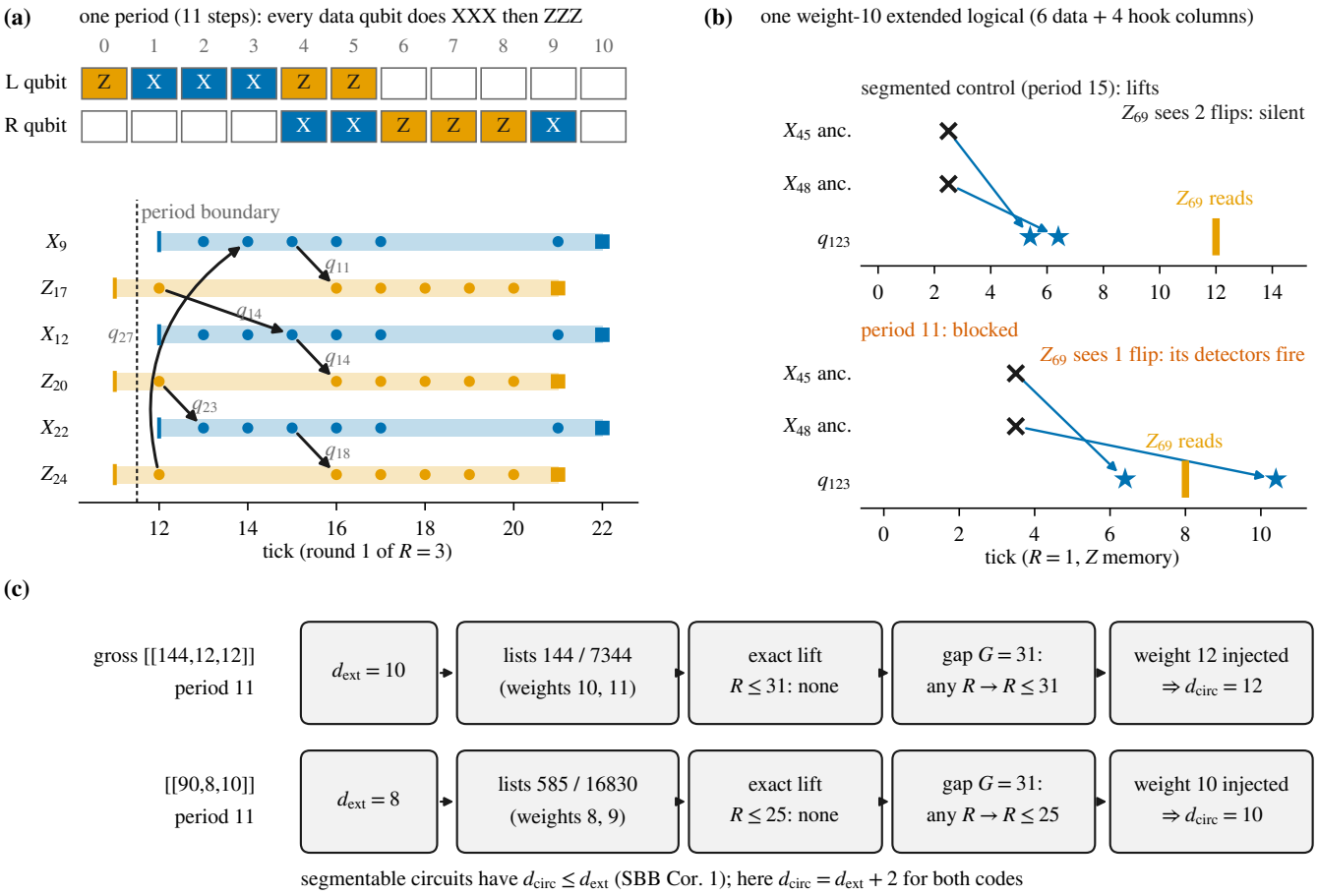


FIG. 1. The period-11 gross circuit. (a) Top: CNOT types on a left and a right data qubit over one period. Bottom: a cycle $X_9 \to Z_{17} \to X_{12} \to Z_{20} \to X_{22} \to Z_{24} \to X_9$ of check instances; each arrow is a shared data qubit on which the first instance acts first. On left data qubits a Z instance's first CNOT precedes the round's X CNOTs; on right data qubits an X instance's last CNOT follows the round's Z CNOTs; so X and Z instances of one round precede one another around cycles, and the circuit is not segmentable. (b) A weight-10 extended logical (Z memory, $R = 1$). In a segmented circuit with the same orders, hooks of X_{45} and X_{48} reach qubit 123 before Z_{69} reads it, and the logical lifts. In the period-11 circuit Z_{69} reads it between the two arrivals, so both of its detectors fire and the logical cannot lift. (c) The proofs of Theorems 1 and 2.

Theorem 2 *Under SD6 noise, the* $[[90, 8, 10]]$ *circuit of Table I has* $d_{\text{circ}} = 10$ *in both memory bases for every number of rounds* $R \geq 1$.

Proof sketch. As for Theorem 1, with $d_{\text{ext}} = 8$ and the lists of 585 and 16830 extended logicals: for $R = 1, \ldots, 25$ and both bases none lifts (two independent chains of exact computations), the gap argument reduces every R to $R \leq 25$, and a weight-10 data logical gives the upper bound (End Matter). □ The same holds for a non-wrapping period-12 circuit and for a period-16 circuit (SM S4). For every assignment of per-check orders the extended code of this code has a logical of weight at most 8 on the X-check side, so every non-interleaved circuit has $d_{\text{circ}} \leq 8$ by SBB's Cor. 1, a value that is attained.

Certificates. Every infeasibility claim used in the proofs of both theorems is accompanied by a checked certificate. For the period-11 gross circuit, $d_{\text{ext}} = 10$ and the completeness of the lists of 144 weight-10 and 7344 weight-11 extended logicals (both bases) are certified by pseudo-Boolean proofs checked with VeriPB, and every lift instance for $R \leq 31$ by a DRAT proof checked with drat-trim and, for its LRAT core, cake_lpr. For the period-11 $[[90, 8, 10]]$ circuit the same holds with $d_{\text{ext}} = 8$, the lists of 585 and 16830 logicals, and lift instances for $R \leq 25$, the bound the argument for every R needs; likewise for the period-16 circuit ($R \leq 12$). The encodings were generated from the circuits by our code and cross-checked against independent implementations; the reductions to these finite problems are the mathematical arguments above, whose premises were verified computationally (SM S6).

Corollary 1 *The Fig. 12 circuit of SBB has $d_{\text{circ}} = 11$ in both bases for every R, as they conjectured.*

This is SBB's Prop. 3 and Cor. 1 combined with our value $d_{\text{ext}} = 11$ for its orders, from two independent exhaustive searches. Retiming Fig. 12 without changing its orders cannot reach 12 (End Matter). An earlier circuit of ours with orders in the class of Fig. 12, which share its extended code, reaches 12 at period 18 (SM S12).

Logical error rates. We sampled both period-11 circuits and the references with identical BP-OSD and Relay-BP decoders and exact priors, rescaling only the idle channel of SD6, $p_{\text{idle}} = \lambda p$ (SM S7); setting idle noise to zero to isolate the effect of CNOT ordering follows SBB's App. G.2 [4]. Without idle errors ($\lambda = 0$) the $[[90, 8, 10]]$ circuit has a lower per-round logical error rate than IBM's depth-8 circuit at every sampled p between 2×10^{-3} and 5×10^{-3} with both decoders (at $p = 3\times10^{-3}$, BP-OSD: 0.52 [0.44, 0.62], pooled from two independent sampling pipelines); at $\lambda = 0.1$, an idle strength comparable to the SI1000 model [17], the two are statistically indistinguishable, and under standard SD6 ($\lambda = 1$) the new circuit is worse by a factor of 6–9 (BP-OSD) and 9–17 (Relay-BP) per round. At $\lambda = 0$ the gross circuit is below SBB's distance-11 circuit at every sampled p between 3×10^{-3} and 5×10^{-3} with BP-OSD (ratios 0.31–0.56) and comparable to IBM's circuit (0.64 [0.48, 0.85] at $p = 3\times10^{-3}$, pooled over two pipelines); at $\lambda = 0.1$ we resolve no difference (intervals spanning factors of 3–10), and at $\lambda = 1$ it is worse than IBM's circuit by 7 [4, 12] (BP-OSD, $p = 2 \times 10^{-3}$) to 21–39 ($p = 3 \times 10^{-3}$). At $\lambda = 0$ its failures are dominated by BP-OSD decoding failures rather than by minimum-weight logicals. The full distance thus pays off when idling is nearly free; otherwise its gain is paid for in idle locations, the trade-off anticipated in SBB's App. F [4].

Neutral atoms. We also use a neutral-atom circuit-level model with literature parameters: CZ error 5×10^{-3} [20, 21], or a projected 3×10^{-3} [22–24], and dephasing of every atom during each array move and mid-circuit readout with $T_2 = 1$ s [21] or an assumed 10 s [13, 25]. Our period-11 circuits need the same number of rigid array moves per round as IBM's depth-8 circuits (12, fixed by the code) and the same two readouts (SM S8). We assume that the X- and Z-ancilla arrays move one after the other [25], that each move takes a fixed time, and that atoms left unpaired during a Rydberg pulse suffer no error, which experiments report only for atoms parked in a separate storage zone [21, 26]. Under these assumptions the $[[90, 8, 10]]$ circuit has a lower per-round logical error rate than IBM's at every sampled setting with both decoders (ratios 0.40–0.82, all 95% intervals below 1), and the gross circuit is comparable to IBM's (0.46–0.99). If instead unpaired atoms suffer half the CZ error and the two arrays move simultaneously, the ordering reverses, by factors of 2.4 ($[[90, 8, 10]]$) and 3.3 (gross) at current parameters with BP-OSD, because the period-11 circuits spread their CNOTs over more pulses and leave more atoms unpaired.

Discussion. SBB's obstruction is specific to non-interleaved circuits: both codes admit single-ancilla circuits that preserve their full distance, two units above the extended-code distance of their orders. Choosing an order class and deciding the lift problem for the complete lists of extended logicals of weight d_{ext} and $d_{\text{ext}} + 1$ turns an open-ended circuit search into a finite, exactly decidable problem that applies to every single-ancilla circuit limited by SBB's Cor. 1. Within translation-invariant $[[90, 8, 10]]$ schedules whose orders lie in a class with $d_{\text{ext}} = 8$, no period below 11 reaches distance 10 in a bulk Pauli-frame model (End Matter). The natural next goal is a shorter period, approaching the 8 steps of the IBM circuits, which would reduce the idle cost. Correlated faults, leakage and CZ-native compilations are outside the scope of this work.

Data availability. The circuit specifications, the lists of extended logicals used in the lift computations, the witnesses, the certificates (clause sets and proofs) and the reproduction scripts are available at [Zenodo DOI].

AI use. This research was carried out with AI agents (Claude, Anthropic). The agents proposed hypotheses, wrote and ran the search and verification code, worked out the constructions and proofs, and independently rechecked each claim reported here, using separately written code for every computation. The author directed the project, made all decisions at its decision points, and takes full responsibility for the content. In line with APS policy, no AI system is listed as an author.

* 3250101086@zju.edu.cn

[1] S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, and T. J. Yoder, High-threshold and low-overhead fault-tolerant quantum memory, Nature **627**, 778 (2024), arXiv:2308.07915.

[2] T. J. Yoder, E. Schoute, P. Rall, E. Pritchett, J. M. Gambetta, A. W. Cross, M. Carroll, and M. E. Beverland, Tour de gross: A modular quantum computer based on bivariate bicycle codes, arXiv:2506.03094 (2025).

[3] L. Aghababaie Beni, O. Higgott, and N. Shutty, Tesseract: A search-based decoder for quantum error correction, arXiv:2503.10988 (v2, 2025).

[4] A. Strikis, D. E. Browne, and M. E. Beverland, High-performance syndrome extraction circuits for quantum codes, arXiv:2603.05481 (v2, 2026).

[5] A. G. Manes and J. Claes, Distance-preserving stabilizer measurements in hypergraph product codes, Quantum **9**, 1618 (2025), arXiv:2308.15520.

[6] M. Webster, A. Jacob, and O. Higgott, Distance-finding algorithms for quantum codes and circuits, arXiv:2603.22532 (v2, 2026).

[7] Z. Liang, B. Gu, Y.-A. Chen, J. Eisert, and Z. Wang, Ultra-high-distance quantum memories from amplified qLDPC codes, arXiv:2609.37231 (2026).

[8] Z. Sun and Z. Cai, Layer codes as quantum memories: syndrome extraction, thresholds and idle robustness, arXiv:2609.39522 (v2, 2026).

[9] M. H. Shaw and B. M. Terhal, Lowering connectivity requirements for bivariate bicycle codes using morphing circuits, Phys. Rev. Lett. **134**, 090602 (2025), arXiv:2407.16336.

[10] A. Wills, M. E. Beverland, L. S. Bishop, J. M. Gambetta, P. Rall, V. Siddhu, and A. W. Cross, Concatenating algebraic codes over high-rate quantum LDPC codes, arXiv:2605.21898 (v2, 2026).

[11] R. Bi, K. Chang, and S. Puri, Untangling QLDPC codes with biased noise ancilla, arXiv:2606.30592 (2026).

[12] M. T. P. Nguyen, M. Rimbach-Russ, and S. Bosco, Disassembling qLDPC codes for depth-optimal parity-check circuits, arXiv:2608.19917 (v2, 2026).

[13] Q. Xu, J. P. Bonilla Ataides, C. A. Pattison, N. Raveendran, D. Bluvstein, J. Wurtz, B. Vasic, M. D. Lukin, L. Jiang, and H. Zhou, Constant-overhead fault-tolerant quantum computation with reconfigurable atom arrays, arXiv:2308.08648 (2023).

[14] Y. Hirai, S. Ikari, Y. Ueno, and Y. Suzuki, No more hooks in the surface code: Distance-preserving syndrome extraction for arbitrary layouts at minimum depth, arXiv:2603.01628 (v2, 2026).

[15] Y. Tomita and K. M. Svore, Low-distance surface codes under realistic quantum noise, Phys. Rev. A **90**, 062320 (2014), arXiv:1404.3747.

[16] G. Kishony and A. Fowler, Color code off-the-hook: avoiding hook errors with a single auxiliary per plaquette, arXiv:2603.28852 (2026).

[17] C. Gidney, M. Newman, A. Fowler, and M. Broughton, A fault-tolerant honeycomb memory, Quantum **5**, 605 (2021), arXiv:2108.10457.

[18] C. Gidney, Stim: a fast stabilizer circuit simulator, Quantum **5**, 497 (2021), arXiv:2103.02202.

[19] See Supplemental Material at [URL will be inserted by publisher] for the proof of Lemma 1 for our template, the period-11 gross circuit, the $[[90, 8, 10]]$ code and its circuits, the direct search on the detector error model, the certificates, the logical error rates, the neutral-atom noise model, the analysis of Fig. 12 and its retimings, an earlier family of period-18 gross circuits, and the verification and data, which includes Refs. [1, 4, 17, 18, 27].

[20] S. J. Evered, D. Bluvstein, M. Kalinowski, S. Ebadi, T. Manovitz, H. Zhou, S. H. Li, A. A. Geim, T. T. Wang, N. Maskara, H. Levine, G. Semeghini, M. Greiner, V. Vuletić, and M. D. Lukin, High-fidelity parallel entangling gates on a neutral atom quantum computer, Nature **622**, 268 (2023), arXiv:2304.05420.

[21] D. Bluvstein *et al.*, Logical quantum processor based on reconfigurable atom arrays, Nature (2023), doi:10.1038/s41586-023-06927-3, arXiv:2312.03982.

[22] J. A. Muniz *et al.*, High-fidelity universal gates in the ^{171}Yb ground state nuclear spin qubit, arXiv:2411.11708 (v2, 2024).

[23] A. G. Radnaev *et al.*, A universal neutral-atom quantum computer with individual optical addressing and non-destructive readout, PRX Quantum **6**, 030334 (2025), arXiv:2408.08288.

[24] R. B.-S. Tsai, X. Sun, A. L. Shaw, R. Finkelstein, and M. Endres, Benchmarking and fidelity response theory of high-fidelity Rydberg entangling gates, PRX Quantum **6**, 010331 (2025), arXiv:2407.20184.

[25] J. Viszlai, W. Yang, S. F. Lin, J. Liu, N. Nottingham, J. M. Baker, and F. T. Chong, qSIEVE: Efficient qLDPC memory via systolic movement in atom arrays, ACM Trans. Quantum Comput. **7**, 9 (2026), arXiv:2311.16980.

[26] D. Bluvstein *et al.*, Architectural mechanisms of a universal fault-tolerant quantum computer, arXiv:2506.20661 (2025).

[27] M. H. Shaw and B. M. Terhal, Optimising quantum error correction using morphing circuits, arXiv:2604.09797 (v2, 2026).

END MATTER

Proof of Lemma 2. Let w faults form an undetected logical, z of them with zero image and c pairs with equal nonzero images, which cancel. By Lemma 1 the remaining distinct columns form an extended logical of weight $w - z - 2c \geq d_{\text{ext}}$. For $w = d_{\text{ext}}$ this forces $z = c = 0$; for $w = d_{\text{ext}} + 1$ it forces $c = 0$ and $z \leq 1$. □

All rounds for Theorem 1. Let $D = 11$ be the period. Each check instance starts at $s_q + jD$ with $s_q \in \{0, 1\}$ ($S = 1$), is measured $\Delta = 10$ layers later, and has its last CNOT at offset 9. If all faults of a set $F_<$ lie at layers $\leq a$, the data frame they produce is constant from layer $a + D - 1$ on, since only ancillas still in flight can change it. A comparison detector responds to both $F_<$ and a set $F_>$ of faults at layers $\geq b$ only if $b - a \leq 2D + \Delta - 2 = 30$; a final detector only if $b - a \leq D + S + \Delta - 1 = 21$; a first-round detector only if $b - a \leq 11$. Hence a fault-free window of $G = 31$ layers splits an undetected logical into two undetected parts whose logical effects add to a nonzero value. With no undetected logical of weight ≤ 9 (Lemma 1), a weight-10 logical spans at most 9×30 layers; with none of weight ≤ 10, a weight-11 logical spans at most $10 \times 30 = 300$. The noisy circuit with R rounds equals the circuit with $R - 1$ rounds with an 11-

layer block inserted at any layer x with $1 \le x \le 11(R-1)$ (layers numbered from 0). Deleting fault-free blocks in front leaves the first fault at a layer ≤ 11, hence the last at ≤ 311; deleting blocks behind is possible while $x = f_{\text{last}}+2D-2 \le 11(R-1)$, so it stops with $11(R-1) < 331$, i.e. $R \le 31$. The exact lifts for $R \le 31$ then exclude every R. □

Details for Theorem 2. The extended code of the orders of Table I equals that of the period-16 circuit, with $d_{\text{ext}} = 8$ and the complete lists of 585 and 16830 logicals. The lifts for $R = 1, \ldots, 25$, both bases, were decided by two independent chains (a frontier dynamic program together with a direct search of the DEM for $R \le 12$, and a separately written SAT lift), all with zero lifts; the gap argument of the previous paragraph, with this circuit's constants, gives $R' = 25$ for weight 9. A weight-10 data logical injected after initialisation fires no detector and flips the observable. The non-wrapping period-12 circuit has separated rounds, $R' = 12$ and zero lifts for $R \le 12$; the period-16 circuit is treated in SM S4.

Family bound for $[[90, 8, 10]]$. Within translation-invariant single-ancilla schedules whose X- and Z-check CNOT orders both lie in one of the four order classes with $d_{\text{ext}} = 8$, no schedule with period $P \le 10$ (wrapping allowed) reaches bulk circuit distance 10 (48 of 48 SAT instances unsatisfiable, in the bulk Pauli-frame model of SM S9); period 11 is attained.

Retimings of Fig. 12. Among periodic schedules of period ≤ 8 keeping the Fig. 12 orders there are four conflict-free ones up to translation, all with injection-verified weight-11 undetected errors for $R = 1, \ldots, 6$. For longer periods, in a bulk Pauli-frame model a data error is seen by a check at the first readout after it, and a logical is liftable in the model if its errors can be timed so that every check sees an even number of them between consecutive readouts. A model lift of a logical with ten data errors and one length-2 residual is a genuine undetected error once the number of rounds is large enough (a one-dimensional Helly argument). Every legal circuit keeping the orders has such an error in each family of Table S5, as shown by SAT encodings restricted to these logicals, each accompanied by a DRAT proof (SM S9).

Supplemental Material

The sections follow the order of the main text. S1 fixes notation and S2 proves Lemma 1; S3 treats the period-11 gross circuit and S4 the $[[90, 8, 10]]$ code; S5 describes the direct search on the detector error model, S6 the certificates and S7 the logical error rates; S8 treats the neutral-atom noise model. S9 and S10 concern Fig. 12 of SBB, and S11–S17 an earlier family of gross circuits with period 18. S18 lists the data, and S19 concerns the morphing end-cycle code.

S1. NOTATION AND NOISE MODEL

Code. The gross code [1] has $\ell = 12$, $m = 6$, $A = x^3 + y + y^2$, $B = y^3 + x + x^2$, $H_X = [A \,|\, B]$, $H_Z = [B^T \,|\, A^T]$; qubits and checks are indexed by $i = 6a + b$ for the monomial $x^a y^b$, with left and right blocks of 72 qubits each; $n = 144$, $k = 12$, $d = 12$. Term labels: $A_1 = x^3$, $A_2 = y$, $A_3 = y^2$, $B_1 = y^3$, $B_2 = x$, $B_3 = x^2$. The X check at (x, y) acts on left qubit $(x, y) + A_k$ and right qubit $(x, y) + B_k$; the Z check at (x, y) on left qubit $(x, y) - B_k$ and right qubit $(x, y) - A_k$. The colour of a check is $(x + y)$ mod 3; the 24 translations (a, b) with $a + b \equiv 0 \pmod 3$ preserve colours.

Circuit template. Each check has one ancilla. In every round it is reset, applies its six CNOTs in its order (ancilla as control for X checks, as target for Z checks) and is measured. A schedule assigns each check a reset step, six CNOT steps and a measurement step modulo the period P; windows may wrap around the period boundary, so that rounds of different checks overlap. A Z-basis memory initialises the data in $|0\rangle$, runs R rounds and reads all data in the Z basis; detectors are the first measurement of each Z check, the comparison of consecutive measurements of each Z check, and the comparison of the last measurement with the parity of the final readout. The X basis is analogous.

Noise and d_{circ}. SD6 [17]: after every CNOT a two-qubit depolarizing channel; after every reset a flip; before every measurement a flip; on every qubit idle in a time step a single-qubit depolarizing channel (also during reset and measurement layers). A fault is one Pauli component of one channel. Faults with identical detector and observable signatures are one DEM mechanism, and d_{circ} counts mechanisms. For lower bounds we use the CSS relaxation: in a Z-basis memory only Z-check detectors and the X part of each fault are kept; this can only lower the minimum. Upper bounds are checked on the full circuit. Lower bounds hold for any sub-model of SD6.

S2. PROOF OF LEMMA 1

We work in a Z-basis memory; the X basis follows by exchanging the roles of X and Z and of the two check types. In a Z-basis memory only X components matter. For a fault we track its *data X frame*: the set s of data qubits carrying an X error after the fault has been propagated through the rest of the circuit to the final data readout. Its *image* is the column $(H_Z s,\, L_Z s)$.

Gates of the template. An X-check ancilla is prepared in $|+\rangle$, is the control of its six CNOTs (data qubits as targets) and is measured in the X basis. A Z-check ancilla is prepared in $|0\rangle$, is the target of its six CNOTs (data qubits as controls) and is measured in the Z basis. Under a CNOT, an X on the control is copied to the target, an X on the target stays there, and Z errors never create X errors. Hence: (a) an X on a data qubit never leaves it (as a target it is not copied; as a control it is copied only onto Z-check ancillas, which are never controls and are measured and reset without acting back on the data); (b) an X on a Z-check ancilla never reaches the data; (c) an X on an X-check ancilla after its k-th CNOT of an instance is copied onto exactly the targets $q_{k+1}, \ldots, q_6$ of the remaining CNOTs of that instance and is then discarded by the next preparation (the X-basis measurement does not see it). These statements use only the direction of each gate, not the relative timing of different checks, so they hold for interleaved and pipelined schedules alike.

Single faults. We go through the SD6 locations. In each case s is empty, a single data qubit, or a suffix $\{q_k, \ldots, q_6\}$ of one X-check order.

1. Data qubits: an X or Y from idle noise, a flip after data preparation, or a flip before the final readout gives $s = \{q\}$ by (a); a Z gives $s = \emptyset$.

2. Z-check ancillas: preparation and measurement flips, idle errors and any error on the ancilla alone give $s = \emptyset$ by (b).

3. X-check ancillas: the preparation flip and the measurement flip are Z errors in this basis and give $s = \emptyset$; an X or Y from idle noise before the first CNOT gives the full support of the check, after the k-th CNOT the suffix $\{q_{k+1}, \ldots, q_6\}$, after the sixth CNOT $s = \emptyset$, by (c).

4. Two-qubit Pauli $P_a \otimes P_d$ after the k-th CNOT of an X check (ancilla a, data q_k): the X part of P_a gives $\{q_{k+1}, \ldots, q_6\}$ and the X part of P_d gives $\{q_k\}$, so $X_a I$ gives $\{q_{k+1}, \ldots, q_6\}$, $I X_d$ gives $\{q_k\}$, and $X_a X_d$ (or any combination with X parts on both) gives their union $\{q_k, q_{k+1}, \ldots, q_6\}$, the suffix starting at q_k.

5. Two-qubit Pauli after a CNOT of a Z check (data q, ancilla): the data part gives $\{q\}$ or $\emptyset$ by (a), the ancilla part gives $\emptyset$ by (b).

Images. A single data qubit gives a data column of the extended code. A suffix $\{q_k, \ldots, q_6\}$ with $2 \le k \le 6$ is a residual, i.e. one of the appended columns (for $k = 6$ it is the data column of q_6). The full support ($k = 1$) is the check itself, whose image is zero because H_Z and L_Z commute with the X stabilisers. So every single fault has image zero or exactly one extended-code column. Merging faults into DEM mechanisms does not change this, since merged faults have identical effects on all detectors and observables, including the augmented ones introduced below.

Telescoping. Augment the circuit with one observable for each row of H_Z and of L_Z, measured on the final data readout. For every Z check i, the sum of all its detectors (first measurement, all consecutive comparisons, final comparison) telescopes to the parity of row i of H_Z on the final readout, i.e. to the augmented observable H_i. Hence an undetected fault set has images whose H parts sum to zero, and whose L parts sum to its logical effect. Its nonzero images therefore sum to an extended-code logical; cancelling equal pairs leaves a set of distinct columns of weight at most the number of faults. If the set has eleven faults and $d_{\text{ext}} = 11$, the eleven images are nonzero, distinct and form one of the listed weight-11 logicals. The template has no X-check detectors in a Z-basis memory; adding them would only remove undetected fault sets, so the bound also holds with all detectors.

Computed check. For every circuit and every R used in S13 we confirmed both facts on the DEM in both bases: each mechanism's per-check detector parity equals the H part of its image, its logical effect equals the L part, and the set of images is exactly the 360 extended columns.

S3. THE PERIOD-11 GROSS CIRCUIT

Circuit. Table I specifies the circuit: every X check is reset at step 0, applies $A_2 A_1 A_3 B_3 B_2 B_1$ at steps 1, 2, 3, 4, 5, 9 and is measured at step 10; every Z check is reset at step 10, applies $B_1 B_2 B_3 A_3 A_1 A_2$ at steps 0, 4, 5, 6, 7, 8 of the next period and is measured at step 9. The 72 translations map the schedule to itself. The circuit is legal (every check's six CNOTs cover its support at distinct offsets inside its window; no data qubit takes part in two CNOTs at one step modulo 11) and deterministic (the noiseless circuit fires no detector; all detectors are deterministic for $R = 1, 2, 3$ in both bases). Our encoder and the benchmark's give identical detector error models for $R = 1, 2, 3$.

Non-segmentability. Within one period the cyclic sequence of CNOT types on a left data qubit is $ZXXXZZ$ (steps 0–5) and on a right data qubit $XXZZZX$ (steps 4–9): three X-check CNOTs followed by three Z-check CNOTs. Consider the precedence graph on check instances, with an edge whenever, on a shared data qubit, a CNOT of one instance precedes a CNOT of an instance of the other type. A segmentable circuit would admit block numbers that never decrease along edges and increase across types. The graph of this circuit has strongly connected components containing both types (144 instances each, all instances of one round label), for example the 6-cycle $X_9 \to Z_{17} \to X_{12} \to Z_{20} \to X_{22} \to Z_{24} \to X_9$, so no such numbering exists. A Z-check instance's CNOTs fall in steps 0 and 4–8 of one period: on left data qubits a Z instance's first CNOT precedes the round's X CNOTs, and on right data qubits an X instance's last CNOT follows the round's Z CNOTs, so X and Z instances of one round precede one another around cycles. No pair of instances precedes each other on two shared qubits; the crossing closes only around cycles. A circuit with the same orders but an X block at steps 1–6 and a Z block at steps 8–13 (period 15) has no mixed component.

Order class and lists. The X order is $A_1 A_2 A_3 B_3 B_1 B_2$ after the swaps s_{12} and s_{56}, and the Z order is the reversal of the X order; both lie in the class of $A_1 A_2 A_3 B_3 B_1 B_2$ generated by reversal, s_{12} and s_{56}, so they share one extended code (S11). It has 360 merged columns per basis (5 orbits under the 72 translations), no logical of weight ≤ 9, and exactly 144 logicals of weight 10 and 7344 of weight 11 in each basis. Two independent exhaustive enumerations (a depth-first search, and a separately written search whose last three levels use a meet-in-the-middle table) give identical sets; neither finds a logical of weight ≤ 11 made of data columns only, which recomputes $d \ge 12$.

Lift. In every round the zero-image mechanisms with detectors are exactly the 72 ancilla readout flips, there are no other zero-image mechanisms, and every DEM mechanism has a unique image. Following Lemma 2, we decide for every weight-10 extended logical whether it lifts, with or without one zero-image mechanism, and for every weight-11 extended logical whether it lifts. For $R = 1, \ldots, 31$ and both bases no instance lifts; the image property and the telescoping identity were asserted on the DEM for every R used. Three implementations agree (a frontier dynamic program, a CP-SAT model and a per-logical SAT model). Positive controls: a period-10 control circuit lifts 432 (Z)

and 576 (X) weight-11 logicals at $R = 1, 2$; the segmented circuit above lifts all 144 weight-10 logicals, 144 with a readout flip and all 7344 of weight 11.

All R and upper bound. The premises of the gap argument of the End Matter were checked on the noisy circuits: start offsets $s_q \in \{0, 1\}$ for every R, measurement offset 10, last CNOT offset 9, and block insertion at every layer $x \in [1, 11(R-1)]$ (layer by layer, both bases, $R \leq 16$ with our encoder and $R \leq 12$ with the benchmark's). Weight-12 data logicals (supports $\{4, 51, 52, 55, 58, 60, 64, 70, 125, 126, 127, 142\}$ of X type and $\{3, 5, 19, 20, 21, 22, 39, 41, 55, 56, 57, 58\}$ of Z type), applied as flips after the data reset, fire no detector and flip the observable for $R = 1, 2, 3, 6, 10, 31$ in both bases.

Search and control. The circuit was found by a lazy-clause SAT search over translation-invariant schedules with orders in this class, blocking the modelled logicals in the bulk Pauli-frame model of S9, and then certified exactly. About one in 70 random legal deterministic schedules of the same family passes the exact screen for $R \leq 2$; the result therefore rests on the exact certification, not on the efficiency of the search.

S4. THE $[[90, 8, 10]]$ CODE: BOUND, LISTS AND CIRCUITS

TABLE S1. The period-16 $[[90, 8, 10]]$ circuit. All ancillas are reset at step 0 and measured at step 15; steps 11, 13 and 14 contain no CNOT; every check of a type uses the same order and steps.

type	CNOT order	CNOT steps
X	$A_2A_1A_3B_2B_3B_1$	1, 3, 5, 6, 8, 10
Z	$B_1B_3B_2A_3A_1A_2$	2, 4, 6, 7, 9, 12

Code and circuit. $\ell = 15$, $m = 3$, $x = S_{15} \otimes I_3$, $y = I_{15} \otimes S_3$, $A = x^9 + y + y^2$, $B = 1 + x^2 + x^7$, $H_X = [A \,|\, B]$, $H_Z = [B^T \,|\, A^T]$ [1]; $n = 90$, $k = 8$, and an exhaustive search finds no logical of weight ≤ 9 and 558 X-type logicals of weight 10, so $d = 10$. Labels: $A_1 = x^9$, $A_2 = y$, $A_3 = y^2$, $B_1 = 1$, $B_2 = x^2$, $B_3 = x^7$; qubits and checks are indexed by $i = 3a + b$, and the check at (x, y) acts as in S1. The 45 translations are code automorphisms that preserve the labels, and the period-16 circuit of Table S1 is invariant under all of them. It is legal (each data qubit takes part in at most one CNOT per step; every check's six CNOTs cover its support at distinct steps inside its window) and deterministic (each of the 405 overlapping pairs of X and Z checks shares an even number of qubits on which the X CNOT precedes the Z CNOT). X-type CNOTs occupy steps $\{1, 3, 5, 6, 8, 10\}$ and Z-type CNOTs steps $\{2, 4, 6, 7, 9, 12\}$, so each data qubit's cyclic sequence is $XZXZXZ$ and no partition of a round into an X part and a Z part exists.

Every choice of orders has $d_{\text{ext}} \leq 8$. Up to s_{12}, s_{56} and reversal, a check of this code has 90 order types; for 36 of them the extended code of that check alone already has a logical of weight ≤ 8. For the remaining 54, consider each of the 558 weight-10 X-type logicals l: any realised residual columns inside l whose total size reduction is at least 2 (one triple or quadruple, or two disjoint pairs) give, with the rest of l, an extended logical of weight ≤ 8. We encode the choice of one admissible type per check as a SAT instance with one clause for each such combination (17910 combinations, each verified explicitly as an extended logical of weight ≤ 8; 3555 variables and 21690 clauses), in a form that only strengthens the statement (types are required "at least one" per check, and realisation is a one-way implication). CaDiCaL reports unsatisfiability and the DRAT proof is verified by drat-trim. Hence for every assignment of orders the X-side extended code has a logical of weight ≤ 8, and every non-interleaved circuit has $d_{\text{circ}} \leq 8$ by SBB's Cor. 1. A non-interleaved circuit whose extended code equals that of Table S1 lifts all 585 weight-8 extended logicals at $R = 1$, so 8 is attained.

Lists. The extended code of the orders of Table S1 has 225 merged columns in each basis (5 orbits under the 45 translations). There are no extended logicals of weight ≤ 7, exactly 585 of weight 8 and 16830 of weight 9 in each basis; a depth-first search and an independent meet-in-the-middle enumeration give identical sets.

Lift. Following Lemma 2, we decide for every weight-8 extended logical whether it lifts, with or without one additional zero-image DEM mechanism, and for every weight-9 extended logical whether it lifts. Zero-image mechanisms with detectors are, in every round, exactly the 45 ancilla readout flips (checked for $R = 1, \ldots, 12$); all zero-image mechanisms of the DEM are admitted as candidates, so this description is not used. For $R = 1, \ldots, 12$ and both bases no instance lifts (full counts, no early stopping), with three implementations: a frontier dynamic program, a CP-SAT model, and a per-logical SAT model; at $R = 1$ an exhaustive enumeration of the products agrees. Controls: a nearby interleaved circuit with $d_{\text{circ}} = 9$ gives 315 weight-9 lifts at $R = 1$ and $R = 2$; the non-interleaved circuit above gives 585 weight-8 lifts, 540 weight-8 lifts with a zero-image fault, and 16830 weight-9 lifts at $R = 1$. The image property

(S2) and the telescoping identity were asserted on the DEM for every R used.

All R. In round j all CNOTs lie in layers $16j+1,\dots,16j+12$ and all measurements in layer $16j+15$, and the noisy circuit with $R+1$ rounds equals the R-round circuit with one 16-layer block inserted (checked for $R=3,\dots,14$, both bases). If a minimal undetected logical has a fault-free round strictly inside its span, the faults before that round form an undetected set by themselves (from that round on every measurement reads the same syndrome, consistent with the final readout), so both parts are undetected and one of them is a lighter logical. Hence a weight-9 logical spans at most 9 rounds, and deleting fault-free rounds after the second and before the last maps it into a circuit with at most $2+9+1=12$ rounds.

Upper bound. A weight-10 data logical, applied as flips after data initialisation, fires no detector and flips the observable for $R=1,2,3,6,10,13$ in both bases.

Period-11 and period-12 circuits. The period-11 circuit of Table I wraps (every Z-check window crosses the period boundary; start offsets $\{0,1\}$, measurement offset 10), and the period-12 circuit does not wrap. Both are legal, deterministic and translation invariant, and the column set of their extended code equals that of the period-16 circuit, so the lists of 585 and 16830 logicals apply. For the period-11 circuit the lifts for $R=1,\dots,25$ are zero in both bases, decided by a frontier dynamic program together with the direct search of S5 for $R\le 12$, and independently by a separately written SAT lift; the image property and the telescoping identity were asserted for every R. Block insertion holds at every layer $x\in[1,11(R-1)]$ (checked layer by layer for $R\le 15$, both bases), and the gap argument gives $R'=22$ for weight 8 and $R'=25$ for weight 9. For the period-12 circuit rounds are separated and $R'=12$; its lifts are zero for $R\le 12$. A weight-10 data logical injected after the data reset fires no detector and flips the observable for $R=1,2,5,11$ in both bases.

Family bound. Within translation-invariant schedules whose X- and Z-check orders both lie in one of the four order classes with $d_{\mathrm{ext}}=8$, every legal deterministic schedule with period $P\le 10$, wrapping allowed, has a liftable weight-≤ 9 logical in the bulk Pauli-frame model (48 of 48 SAT instances unsatisfiable, re-run spot checks agree). The statement concerns many rounds and this family only: order classes with $d_{\mathrm{ext}}\le 7$, smaller symmetry and per-check-free schedules are not covered.

S5. DIRECT SEARCH ON THE DETECTOR ERROR MODEL

The search looks for a minimum-weight set of DEM mechanisms with zero detector sum and nonzero logical effect. It branches on the mechanisms covering an odd detector, excluding earlier siblings so that the branches partition the search space; it prunes any partial set whose detector sum and logical effect both vanish (such a set cannot be contained in a minimal solution), bounds the remaining weight by a packing of detectors with disjoint covering mechanisms, and closes the last level by table look-up. Root cases are split by the orbits of the translations that preserve the detector set and the logical classes (checked by rank conditions). This is the same family of methods as the extended-code search of S10, implemented independently with different pruning; its $R=1$ results were reproduced by a second, independently written implementation, which also reproduced $d_{\mathrm{ext}}=11$ and the list of 4224 logicals of S10, and both were tested on 1672 random instances with known answers. The searches produce no proof certificates.

Results. Gross circuit A: at $R=1$ there is no undetected logical of weight ≤ 11 in either basis and one of weight 12 is found; at $R=2$ (66 root cases) and $R=3$ (96 root cases) there is none of weight ≤ 11, and the injection of S15 gives 12. $[[90,8,10]]$ circuit: for $R=1,\dots,12$ there is none of weight ≤ 9 in either basis, and weight 10 is found for $R=1,2,3$; the interleaved control circuit gives 9. For the $[[90,8,10]]$ circuit, take a minimal undetected logical of weight $w\le 9$ at any R: by the splitting argument of S4 it spans at most w rounds and maps into a circuit with at most $w+3\le 12$ rounds, contradicting the search. This proves $d_{\mathrm{circ}}=10$ for the period-16 circuit for every R without extended-code input; it uses the separation and periodicity checks of S4. At $R=1$ a SAT formulation of the same question on the DEM is also unsatisfiable in both bases.

Period-11 circuits. For the period-11 gross circuit, the search (in the second, independently written implementation, whose last levels use a meet-in-the-middle table) finds no undetected logical of weight ≤ 11 at $R=1$ and $R=2$ in both bases; a period-10 control circuit gives 11. For the period-11 $[[90,8,10]]$ circuit there is no undetected logical of weight ≤ 9 for $R\le 12$.

S6. CERTIFICATES

A VeriPB proof is accepted only with the verdict "s VERIFIED UNSATISFIABLE"; DRAT proofs are checked with drat-trim and their LRAT cores with the formally verified checker cake_lpr; lift instances are combined into

batches by a selector disjunction, so that unsatisfiability of a batch implies that of each member.

Period-16 $[[90, 8, 10]]$ *circuit.* Every infeasibility claim used in the proof is accompanied by a checked certificate: $d_{\text{ext}} \geq 8$ and the completeness of the weight-8 and weight-9 extended-logical lists by VeriPB (including the reduction to orbit representatives, which is part of the checked proof), and all lift instances for $R \leq 12$, both bases, by DRAT proofs. $d_{\text{ext}} \leq 8$ is witnessed by the listed logicals.

Period-11 gross circuit. $d_{\text{ext}} \geq 10$ and the completeness of the lists of 144 weight-10 and 7344 weight-11 extended logicals, both bases, are certified by VeriPB 3 proofs (the listed logicals witness $d_{\text{ext}} \leq 10$); all lift instances for $R \leq 31$, both bases, are certified by DRAT proofs. The largest proofs (4.6×10^{11} bytes) are generated deterministically and checked as streams; we record the SHA-256 hash of each stream so that it can be regenerated and compared. Checking the weight-11 proofs requires about 18 GB of memory.

Period-11 and period-12 $[[90, 8, 10]]$ *circuits.* The period-11 circuit uses the extended code of the period-16 circuit, so $d_{\text{ext}} \geq 8$ and the completeness of the weight-8 and weight-9 lists are certified as above; all lift instances for $R \leq 25$, both bases, the bound the argument for every R needs, are certified by DRAT proofs. For the period-12 circuit all lift instances for $R \leq 12$ are certified.

Fig. 12 and the period-18 circuits. $d_{\text{ext}} \geq 9$ for the Fig. 12 order class is certified; $d_{\text{ext}} = 11$ and the list of 4224 weight-11 logicals rest on two independent exhaustive searches. The certificates for the period-18 circuits and for Table S5 are listed in S18.

The encodings were generated from the circuits by our code and cross-checked against independent implementations; the reductions to these finite problems (Lemma 1, the structure of light logicals, and the bounds on R) are mathematical arguments whose premises were verified computationally.

S7. LOGICAL ERROR RATES OF THE PERIOD-11 CIRCUITS

Setup. The circuits and the references (IBM's depth-8 circuits and Fig. 12) are built from the same template and noised by the same SD6 implementation, with the idle channel multiplied by λ; both bases are sampled separately and added. BP-OSD (min-sum, 10000 iterations, OSD-CS order 7) and Relay-BP use identical parameters and exact priors for all circuits.

$[[90, 8, 10]]$. At $\lambda = 0$ the period-11 circuit has a lower per-round logical error rate than IBM's depth-8 circuit at every sampled p between 2×10^{-3} and 5×10^{-3} with both decoders. At $p = 3 \times 10^{-3}$ with BP-OSD an independent sampling pipeline gives 0.57 [0.47, 0.70] and the pooled value is 0.52 [0.44, 0.62]; the data do not show a clear dependence of the ratio on p. At $\lambda = 0.1$ the two circuits are statistically indistinguishable, and at $\lambda = 1$ the period-11 circuit is worse by a factor of 6–9 with BP-OSD and 9–17 with Relay-BP per round. At $\lambda = 0$ the period costs nothing, so this comparison isolates the structural gain from $d_{\text{circ}} = 8$ to 10. Leading order: IBM's circuit has 315 (Z) and 540 (X) undetected logicals of weight 8 per round, the period-11 circuit 24507 of weight 10 per round in each basis; an ideal-decoder estimate places the break-even near $p \approx 2 \times 10^{-3}$ for $\lambda = 0$ and 3×10^{-5} for $\lambda = 1$, which is indicative only.

Gross code. At $\lambda = 0$ the period-11 circuit is below Fig. 12 at every sampled p between 3×10^{-3} and 5×10^{-3} with BP-OSD (ratios 0.31–0.56; at 3×10^{-3}, 0.33 [0.23, 0.49] in an independent pipeline) and at 4–5×10^{-3} with Relay-BP (0.35–0.54). Against IBM's depth-8 circuit it is comparable: BP-OSD point estimates are 0.64–0.96 between 3×10^{-3} and 5×10^{-3}, none individually conclusive (at 3×10^{-3}: 0.73 [0.50, 1.05] from an independent pipeline with a pre-set sample size; 0.64 [0.48, 0.85] pooled over both pipelines). At $\lambda = 0.1$ we resolve no difference from either reference (95% intervals spanning factors of 3–10), and at $\lambda = 1$ it is worse than IBM's circuit per round by 7 [4, 12] (BP-OSD, $p = 2 \times 10^{-3}$) to 21–39 ($p = 3 \times 10^{-3}$, BP-OSD and Relay-BP). At $\lambda = 0$ and $p = 4 \times 10^{-3}$, all failures in a sample of 8000 shots per basis occurred in shots where belief propagation did not converge, and in all but one the OSD output had a higher cost than the actual error: logical error rates in this regime are dominated by decoder failures rather than by minimum-weight logicals.

IBM's gross circuit. Exact enumeration on the detector error model finds exactly 144 (Z basis) and 288 (X basis) undetected logicals of weight 10 per round, constant for $R = 1$–3, and none of lower weight for $R \leq 3$: its circuit distance is 10 for up to three rounds. For the period-11 circuit we find 69756 logicals of weight 12 at $R = 1$ in each basis and 233244 at $R = 2$ (Z basis); the per-round count has not converged.

S8. LOGICAL ERROR RATES IN A NEUTRAL-ATOM NOISE MODEL

Architecture and timing. Data atoms stay in the entangling zone; the X- and Z-ancilla arrays are held in movable tweezers and translated rigidly, and each translation is followed by a global Rydberg CZ pulse and local single-qubit gates (CX = H·CZ·H on the target). One move serves one pair (check type, polynomial term). Both codes have three terms in each of A and B, so every translation-invariant schedule, of any period, needs exactly 12 moves per round; this holds for the period-11 circuits and for IBM's depth-8 circuits, which also have two mid-circuit readout layers per round. The period only determines how the 12 moves are distributed over layers. In the main model the two ancilla arrays move one after the other, as in qSIEVE [25], and each move takes a fixed time, so the modelled round times agree. Equal move counts do not imply equal durations: with a move time growing as the square root of the distance, the summed path is 6% longer for the period-11 gross circuit and 21% longer for the period-11 $[[90, 8, 10]]$ circuit (an indicator that depends on the layout); at current parameters this changes the fault mass by less than 2%. Wrapping around the torus needs two steps per term in qSIEVE [25], so 12 is a lower bound; this applies to both circuits equally.

Noise. Before each layer every atom receives a Z error with probability $(1 - e^{-\tau/T_2})/2$, with τ the move time plus the readout wait; during a mid-circuit readout the unmeasured atoms receive the scattering error of Table S2; preparation, gates, measurement and the two ancilla transfers per round carry the errors of Table S2. The per-round fault mass of each period-11 circuit agrees with IBM's within 0.1% (gross, X basis, current parameters: 5.855 vs. 5.853). The set "cur×0.5" halves every error probability of "cur"; it is a uniform scaling, not a literature value.

TABLE S2. Parameters of the neutral-atom model (error probabilities unless stated). "cur": current hardware; "proj": projected.

quantity	cur	proj	sources and remarks
CZ	5.0×10^{-3}	3.0×10^{-3}	cur: 99.5% [20], [21, 26]; proj: 99.72%, 99.73% with loss excluded [22, 23], and 99.71% [24] (leakage-corrected, from a different architecture)
single-qubit gate	9×10^{-4}	3×10^{-4}	cur: local gates, 99.912% [21], 99.902% [23]; proj: global Raman estimate 99.97% [20], applied here to local gates
preparation	6.8×10^{-3}	4×10^{-3}	99.32% [21]; 99.6% [20]
measurement	5.8×10^{-3}	2×10^{-3}	0.46% flips plus half of 0.24% loss [26]; imaging ≈99.8% [21]
transfer	1×10^{-3}	1×10^{-3}	$\lesssim$0.1% per transfer [21]
T_2	1 s	10 s	cur [21, 26]; proj is an assumption, as in [13, 25]
time per move	200 μs	200 μs	[21]
readout wait	1 ms	0	cur [21]; proj: pipelined with the next moves [13], assumes a second ancilla set
readout scattering	6.8×10^{-4}	3.4×10^{-4}	0.034% per 500 μs image [21], applied for 1 ms

Sampling. The gross circuits are sampled for $R = 12$ and the $[[90, 8, 10]]$ circuits for $R = 10$, both bases separately, with BP-OSD and Relay-BP as in S7. Shot numbers were fixed in advance for each block of circuits; no block was extended after its failures were seen. Intervals are Jeffreys intervals per basis, combined by sampling. An independent implementation of the model gives identical detector error models for all circuits and parameter sets, and independent sampling reproduces the ratios (e.g. $[[90, 8, 10]]$, proj, BP-OSD: 0.575 [0.465, 0.710]).

For the $[[90, 8, 10]]$ code all six intervals lie below 1. For the gross code all six point estimates lie below 1 but only one interval does (cur×0.5, BP-OSD), so we call the two circuits comparable.

Sensitivity. The CZ value 6.5×10^{-3} corresponds to the fidelities before loss is excluded [22, 23]. In cur_slow each gate layer takes 470 μs and each readout layer 2.57 ms, the split of the 4.45 ms round of [26], where 2.57 ms is the time to move the ancillas to the storage zone and bring in the next group. At current parameters the $[[90, 8, 10]]$ advantage is not significant under simultaneous moves (cur_par) or with the measured round (cur_slow); with projected gate errors it persists under simultaneous moves. The last variant is the decisive one. With simultaneous moves IBM's circuits pair almost every atom in every pulse, while the period-11 circuits spread their CNOTs over 10 pulses instead of 7. Counting, for each Rydberg pulse, the atoms that stay in the entangling zone, are neither prepared nor measured in that layer and have no partner, gives 1008 vs. 144 exposures per round for the gross code and 630 vs. 90 for $[[90, 8, 10]]$. If such atoms suffer half the CZ error, the ordering reverses. With sequential moves the counts are equal (1584 and 990 for both circuits). We know of no measurement of this error: experiments report no error on atoms in a separate storage zone [21, 26], which is isolated from the Rydberg beams.

TABLE S3. Per-round logical error rate ratio, period-11 circuit over IBM's depth-8 circuit, with 95% intervals; failures summed over both bases (period-11 / IBM) and shots per basis. Main model (sequential moves, no error on unpaired atoms).

		gross			$[[90, 8, 10]]$		
set	decoder	ratio	failures	shots	ratio	failures	shots
cur	BP-OSD	0.99 [0.81, 1.21]	177 / 179	440	0.82 [0.69, 0.97]	232 / 275	640
cur	Relay-BP	0.80 [0.57, 1.12]	59 / 74	300	0.72 [0.55, 0.95]	87 / 119	530
cur×0.5	BP-OSD	0.46 [0.27, 0.80]	18 / 39	18000	0.54 [0.42, 0.69]	93 / 173	21000
cur×0.5	Relay-BP	0.71 [0.24, 2.2]	5 / 7	25000	0.40 [0.28, 0.56]	44 / 111	48000
proj	BP-OSD	0.70 [0.43, 1.16]	26 / 37	20000	0.57 [0.46, 0.71]	125 / 219	25000
proj	Relay-BP	0.56 [0.20, 1.6]	5 / 9	29000	0.48 [0.35, 0.66]	53 / 111	41000

TABLE S4. Sensitivity variants, BP-OSD; ratio period-11 / IBM with failures (period-11 / IBM).

variant	gross	failures	$[[90, 8, 10]]$	failures
cur, CZ error 6.5×10^{-3}	0.95 [0.81, 1.11]	288 / 305	0.85 [0.71, 1.00]	236 / 260
cur, simultaneous moves (cur_par)	0.98 [0.82, 1.15]	260 / 267	0.89 [0.76, 1.04]	288 / 325
cur, measured round of [26] (cur_slow)	0.92 [0.79, 1.08]	323 / 296	0.92 [0.80, 1.06]	361 / 387
proj, no readout wait, $T_2 = 1$ s, simultaneous moves	0.77 [0.54, 1.08]	56 / 73	0.61 [0.49, 0.76]	133 / 218
cur_par, unpaired atoms with error $p_{\mathrm{CZ}}/2$	3.31 [2.85, 3.84]	603 / 246	2.41 [2.09, 2.77]	561 / 284

Omitted effects and caveats. Atom loss is folded into Pauli errors and the decoders receive no erasure information; T_1 decay and heating are not modelled; the time per move is fixed rather than distance-dependent. At current parameters the gross circuits fail in about 31% of X-basis experiments of 12 rounds (74% for the period-11 circuit in the last variant); in this near-threshold regime the conversion to a per-round rate compresses ratios towards 1, so the current-parameter ratios, including the reversal factors, are conservative. The comparisons hold for these two decoders.

S9. FIG. 12 OF SBB: $d_{\mathrm{circ}} = 11$ AND RETIMINGS

$d_{\mathrm{circ}} = 11$. Fig. 12 is non-interleaved, so SBB's Prop. 3 and Cor. 1 give $d_{\mathrm{circ}} = d_{\mathrm{ext}}$ for every R; $d_{\mathrm{ext}} = 11$ by S10. The weight-11 extended logicals are realised by eleven circuit faults (checked by injection for $R = 1, 2, 3$ in both bases).

Lift structure. All 4224 weight-11 extended logicals per basis lift to undetected circuit errors of Fig. 12 ($R = 1, 2, 3$).

Retimings with period ≤ 8. Consider periodic schedules with period $P \leq 8$ shared by all checks, one reset, six CNOTs at distinct offsets and one measurement per check and period, and no data qubit in two CNOTs at one step. Weight 6 forces $P = 8$ and all offsets; keeping the Fig. 12 orders leaves one start per check. A backtracking search without symmetry assumptions finds exactly 32 conflict-free assignments, 4 up to global translation (two SAT encodings agree); all four are colour-uniform, deterministic and non-interleaved, and one is Fig. 12 itself. For each of them and every $R = 1, \ldots, 6$ in both bases we injected a weight-11 error that fires no detector and flips the observable.

Longer periods (bulk). For longer periods we use a Pauli-frame model of the bulk. A data X error at time θ on qubit w is first seen by Z check c (readouts at $T_c + nP$) in instance $H_c = \min\{n : T_c + nP > \theta\}$, i.e. for $\theta \in (T_c + (H_c - 1)P, T_c + H_c P)$. A weight-11 logical is *model-liftable* if instance indices H_c can be chosen such that the errors seen by each check can be paired, both errors of a pair being seen in the same instance (so all comparison detectors cancel), and, for every data error, the intervals of its three checks have a common point. For three open intervals of length P on the line, pairwise intersection, which is a difference constraint $H_c - H_d \leq \lfloor (T_d - T_c)/P \rfloor + 1$, implies a common point (one-dimensional Helly); an SD6 fault on w is available there (the two-qubit channel after the CNOT with the latest left endpoint contains $X_w \otimes I$). A residual of length 2 is deposited by an ancilla fault after the fourth CNOT of its check, which fixes its arrival times. *Direction used.* If the difference constraints of a logical of ten data errors and one length-2 residual are feasible, the corresponding eleven faults, placed away from the time boundaries, form a genuine undetected logical error; this was derived independently twice and confirmed by 75 direct injections, each firing no detector and flipping the observable. Non-existence statements need exactly this

direction: an unsatisfiable search for a schedule in which no logical is model-liftable shows that every schedule has a model-liftable weight-11 logical, and when the search uses only logicals of the proven form, every schedule has a genuine weight-11 undetected error for sufficiently many rounds. The converse (model blocking implies exact blocking) is not used and fails at the initial time boundary, where the exact lift can succeed via first-round detectors. Blocking a model-liftable logical requires a negative cycle in its constraint graph; we encode this for all modelled logicals, for CP-SAT and in a lazy-clause SAT loop. The certificates block every model-liftable logical. Re-running the same encoders with only the 360 logicals per basis of the form ten data columns plus one length-2 residual gives UNSAT for every row of Table S5, each accompanied by a DRAT proof, so each obstruction rests only on the direction proven above. The resulting status for retimings that keep the Fig. 12 orders is given in Table S5. Per-check-free pipelined schedules with $P \geq 9$ and different CNOT orders are not covered; the latter is how the new circuits escape.

TABLE S5. Retimings of Fig. 12 that keep its CNOT orders. "Obstructed": no legal deterministic circuit of the family blocks every model-liftable weight-11 logical. Each row was also re-solved with only the ten-data-plus-one-length-2-residual logicals, which again gives UNSAT; every obstruction is therefore proven.

family	status
24-translation symmetric, common frame, any P	obstructed
24-translation symmetric, pipelined, $P = 8, \ldots, 12$	obstructed
subgroup-invariant pipelined, $P = 9$: three subgroups of order 12, three of order 6, one of order 8, two of order 4 and one of order 3	obstructed
subgroup-invariant pipelined, $P = 10$: one subgroup each of order 12, 8, 6	obstructed
no symmetry, common frame, $P \leq 16$	obstructed (one encoding)
other pipelined families with $P \geq 9$; symmetric pipelined $P \geq 13$	open

S10. THE EXTENDED CODE: $d_{\text{ext}} = 11$ AND THE 4224 WEIGHT-11 LOGICALS

The extended code of Fig. 12 has 360 merged columns per basis (144 of check-weight 3, 144 of weight 6, 72 of weight 9). The 24 colour-preserving translations map each check's order to the order of its image check and preserve the logical classes (checked by solving $L \circ T = ML + NH$ with $\operatorname{rank} M = k$); the columns form 15 orbits.

Search. To decide whether $Hx = 0$, $Lx \neq 0$ has a solution with $|x| \leq W$ we use a complete depth-first search: a minimum-weight solution may be translated so that it contains the representative of its lowest orbit; the search branches on the columns covering an unsatisfied check row, excluding earlier siblings, so that the branches partition the search space; it prunes when the depth plus $\lceil |s|/w_{\max} \rceil$ exceeds W (s the current syndrome, $w_{\max}$ the largest column weight) and closes the last step by table look-up. The program was validated against unpruned meet-in-the-middle enumeration on smaller instances and by checking that split runs add up to unsplit runs.

Results. At $W = 10$ there is no solution in either basis ($\approx 2 \times 10^{10}$ nodes per basis), on two independently constructed instances, and under a second symmetry group of order 8 (45 orbits, $\approx 5 \times 10^{10}$ nodes per basis); weight 11 is attained. Hence $d_{\text{ext}} = 11$. At $W = 11$, two different search programs enumerate the same 4224 extended logicals per basis, all in free orbits of the 24-element group (176 orbits); 360 of them consist of ten data columns and one residual of length 2. Both $d_{\text{ext}} = 11$ and the completeness of the list of 4224 weight-11 logicals rest on two independent exhaustive searches; we did not obtain a checkable proof certificate for them (a direct CNF encoding is beyond CDCL solvers already at weight 6).

S11. INVARIANCE OF THE EXTENDED CODE

Consider a Z-basis memory; only X errors matter. An X error on the ancilla of an X check after its k-th CNOT propagates to the suffix $\{q_{k+1}, \ldots, q_6\}$; Z-check CNOTs (data controls, ancilla target) do not change the X frame on the data. Each fault's final data X frame is therefore zero, a single data qubit, or a proper suffix of an X-check order, and its column is $(H_Z s, L_Z s)$. Multiplying by the check itself, which has zero column, a suffix of length 5 is equivalent to $\{q_1\}$ (a data column), of length 4 to $\{q_1, q_2\}$, of length 3 to the complement of $\{q_1, q_2, q_3\}$, and a suffix of length 2 is $\{q_5, q_6\}$. Hence the column set is determined by the three sets $\{q_1, q_2\}$, $\{q_1, q_2, q_3\}$, $\{q_5, q_6\}$ of every check. Reversal maps these to $\{q_5, q_6\}$, the last three and $\{q_1, q_2\}$, which produce the same columns, and s_{12}, s_{56} leave them

unchanged. The X basis is symmetric. For circuits A, B and C we also computed the extended codes directly: both bases have 360 merged columns, and the column sets coincide with those of Fig. 12.

S12. EARLIER GROSS CIRCUITS WITH ORDERS IN THE CLASS OF FIG. 12

Before the period-11 circuit we found three circuits whose orders lie in the class of Fig. 12, so that their extended code is that of Fig. 12 ($d_{\text{ext}} = 11$, 4224 logicals of weight 11 per basis): circuit A (period 18, pipelined; Table S6), circuit B (period 18) and circuit C (period 21, all ancillas sharing one frame). Each has $d_{\text{circ}} = 12$ in both bases for every R: by Lemmas 1 and 2 only the 4224 weight-11 logicals need to be excluded, which the lift computation of S13 does for $R \leq 32$ (A), $R \leq 24$ (B) and $R \leq 14$, 16 (C); the round reductions of S14 cover every R, and S15 gives the upper bound. Fig. S1 shows how reversing orders within the class blocks the lifts.

TABLE S6. Circuit A (period 18). Checks are grouped by type and colour $(x + y)$ mod 3; all 24 checks of a class share the entries; steps are modulo 18, and a window passing step 17 continues in the next period; last column: map from the Fig. 12 order of the class (rev: reversal; s_{12}, s_{56}: swap positions 1,2 or 5,6).

class	reset	CNOT order	CNOT steps	meas.	map
X_0	17	$A_1A_2A_3B_3B_1B_2$	0, 1, 2, 8, 9, 15	16	rev s_{56}
X_1	15	$A_1A_2A_3B_1B_3B_2$	16, 3, 4, 5, 10, 13	14	rev
X_2	17	$A_3A_1A_2B_3B_1B_2$	0, 1, 5, 6, 7, 15	16	rev s_{56}
Z_0	17	$B_2B_1B_3A_2A_1A_3$	0, 7, 8, 9, 13, 14	15	rev s_{56}
Z_1	2	$B_2B_3B_1A_3A_2A_1$	3, 4, 9, 10, 11, 17	0	rev $s_{12}s_{56}$
Z_2	16	$B_2B_1B_3A_3A_2A_1$	17, 5, 6, 12, 13, 14	15	rev s_{56}

Table S6 specifies circuit A completely: each of the 144 checks takes the reset step, order, CNOT steps and measurement step of its class. Its resets occur at steps $\{2, 15, 16, 17\}$ and its measurements at $\{0, 14, 15, 16\}$. We verified for circuits A, B and C that every check applies exactly six CNOTs to its support at distinct steps inside its window, that no data qubit takes part in two CNOTs at the same step, that the noiseless circuit fires no detector, and that all detectors are deterministic, for several R in both bases. Interleaving: in circuit A, over one period, 96 data qubits switch four times and 48 twice (cyclically) between X- and Z-type CNOTs.

Circuit B (Table S7) also has period 18; 24 of its checks keep the Fig. 12 order. Circuit C (Table S8) has period 21 and a common frame: all ancillas are reset at step 0 and measured at step 20, so rounds are separated in time. Both have $d_{\text{circ}} = 12$ for every R in both bases (S13, S14).

For reference, the Fig. 12 orders in the same labelling are X_0: $B_1B_2B_3A_3A_2A_1$, X_1: $B_2B_3B_1A_3A_2A_1$, X_2: $B_1B_2B_3A_2A_1A_3$, Z_0: $A_1A_3A_2B_3B_1B_2$, Z_1: $A_2A_1A_3B_1B_2B_3$, Z_2: $A_2A_1A_3B_3B_1B_2$ (period 8, pipelined, non-interleaved).

TABLE S7. Circuit B (period 18). Columns as in Table S6.

class	reset	CNOT order	CNOT steps	meas.	map
X_0	0	$B_2B_1B_3A_3A_2A_1$	1, 7, 8, 14, 15, 16	17	s_{12}
X_1	2	$B_2B_3B_1A_3A_2A_1$	3, 6, 11, 12, 13, 0	1	id
X_2	0	$B_2B_1B_3A_2A_1A_3$	1, 9, 10, 11, 15, 16	17	s_{12}
Z_0	1	$A_3A_1A_2B_3B_1B_2$	2, 3, 7, 8, 9, 17	0	s_{12}
Z_1	17	$A_1A_2A_3B_1B_3B_2$	0, 5, 6, 7, 12, 15	16	$s_{12}s_{56}$
Z_2	1	$A_1A_2A_3B_3B_1B_2$	2, 3, 4, 10, 11, 17	0	s_{12}

S13. LIFT COMPUTATION FOR CIRCUITS A, B AND C

By S2, a weight-11 undetected logical selects, for each of the eleven columns of one listed extended logical, one DEM mechanism with that image, such that the detector sums vanish. For each of the 4224 logicals per basis, a CP-SAT

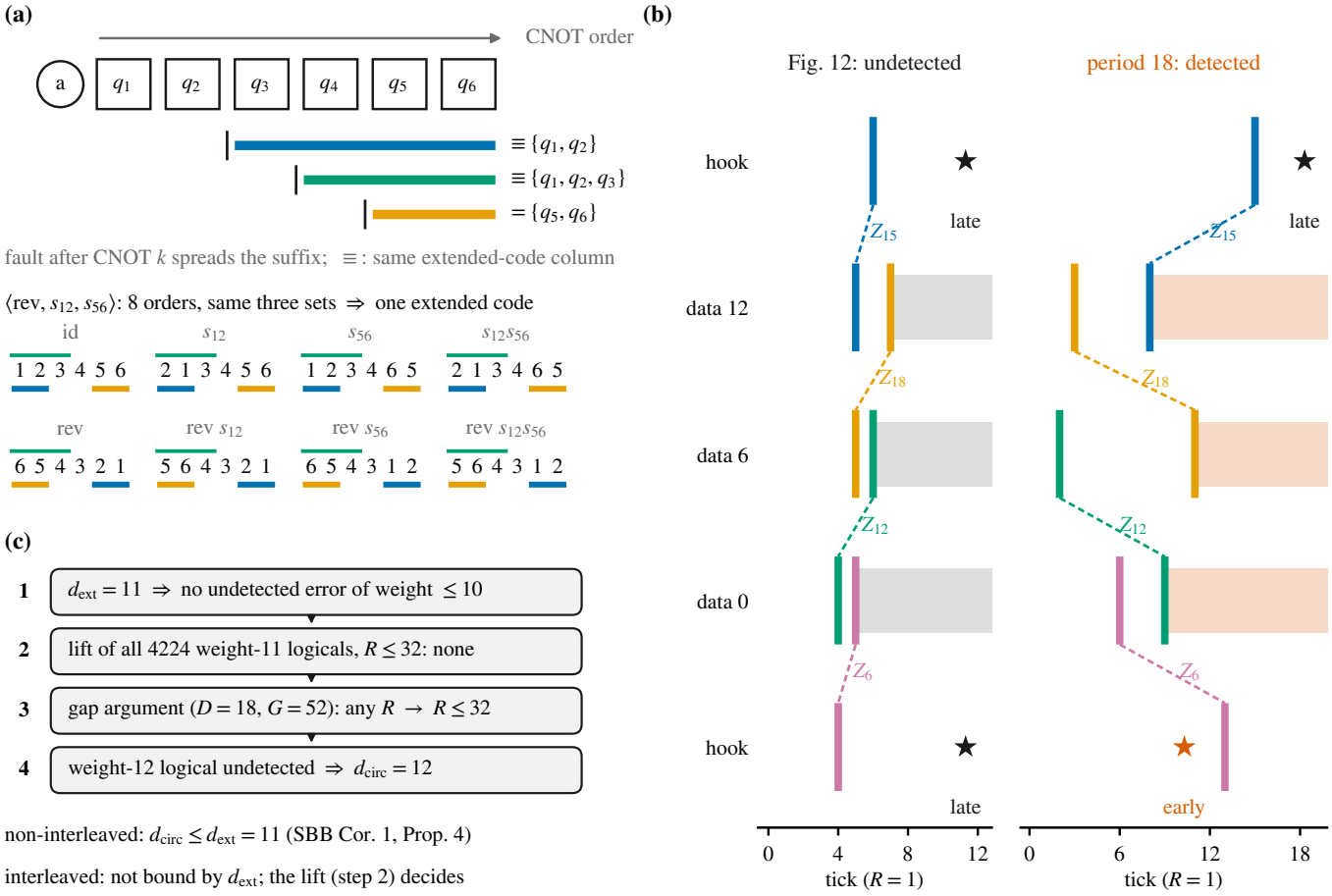


FIG. S1. The idea, on the gross code. (a) Residuals of length 4, 3 and 2 give the same extended-code columns as $\{q_1, q_2\}$, $\{q_1, q_2, q_3\}$ and $\{q_5, q_6\}$, so the eight orders generated by reversal and the swaps s_{12}, s_{56} share one extended code. (b) A weight-11 extended logical (ten data errors and the hook column $\{9, 25\}$; Z memory, $R = 1$). Bars: reads of the Z checks Z_{15}, Z_{18}, Z_{12}, Z_6; stars: hook arrival. Left, Fig. 12: the hook lands after both reads and no detector fires. Right, circuit A: the same column comes from the spread $\{8, 91, 82, 85\}$ (equal to $\{9, 25\}$ up to the check), seen late by Z_{15} and early by Z_6; the forced timings (red) leave odd parity at Z_6. (c) The proof for circuit A.

TABLE S8. Circuit C (period 21, common frame: reset at 0, measurement at 20).

class	CNOT order	CNOT steps	map
X_0	$B_2B_1B_3A_3A_2A_1$	2, 7, 9, 15, 16, 17	s_{12}
X_1	$B_2B_3B_1A_3A_2A_1$	1, 8, 12, 13, 14, 17	id
X_2	$B_2B_1B_3A_2A_1A_3$	1, 10, 11, 12, 16, 18	s_{12}
Z_0	$A_3A_1A_2B_3B_1B_2$	3, 4, 8, 9, 10, 17	s_{12}
Z_1	$A_1A_2A_3B_1B_3B_2$	1, 6, 7, 8, 13, 14	$s_{12}s_{56}$
Z_2	$A_1A_2A_3B_3B_1B_2$	2, 4, 5, 11, 12, 19	s_{12}

model has a Boolean variable per mechanism of the eleven image groups, an exactly-one constraint per group, and an XOR constraint per detector. Infeasibility for all 4224 logicals at a given R means that no weight-11 undetected logical exists at that R.

Results: circuit A, $R = 1, \ldots, 32$, both bases: all infeasible. A second, independently written lift program agrees and extends the range to $R = 36$. Circuit B: $R = 1, \ldots, 24$; circuit C: $R = 1, \ldots, 14$ and 16, plus a solver-free meet-in-the-middle check (split the eleven groups in two halves, enumerate all detector sums of each half, intersect) for $R \leq 3$; all infeasible. Positive control: on Fig. 12 the same programs find a realisation for all 4224 logicals.

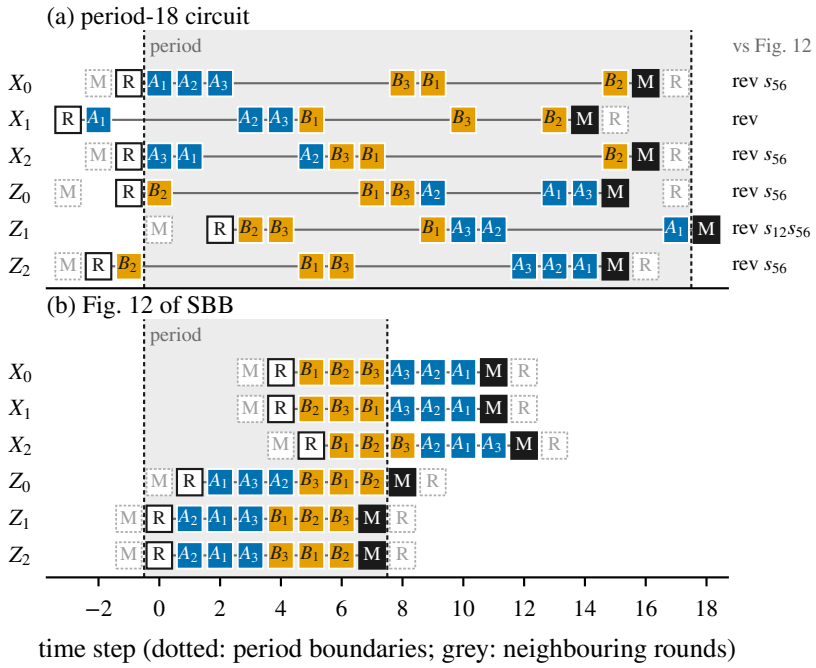


FIG. S2. One period of (a) circuit A of Table S6 and (b) Fig. 12 of SBB [4], on the same time scale. Rows: the six classes; boxes: CNOTs labelled by the term they implement; R, M: ancilla reset and measurement. Windows crossing the period boundary (dotted) continue into the next period. (b) has period 8 and is not interleaved: on every data qubit the X- and Z-type CNOTs of a round form two contiguous blocks.

Certificates. For circuits A ($R \le 32$), B ($R \le 24$) and C ($R \le 14$ and 16), both bases and all 4224 logicals, the lift instances were encoded as CNF. Instances were combined into batches by a selector disjunction: the batch formula is the disjunction of its members, each guarded by its own selector variable, with no variables shared between members, so that unsatisfiability of a batch implies that of each member. Each batch was refuted by CaDiCaL; each refutation is a DRAT proof verified with drat-trim, whose LRAT core was verified with the formally verified checker cake_lpr. As a positive control, every one of the 4224 Fig. 12 instances at $R = 1$ is satisfiable and decodes to a valid lift.

S14. ROUND REDUCTION FOR CIRCUITS A, B AND C

Notation: period $D = 18$; check q has start s_q, instance j is reset at $s_q + jD$ and measured at $s_q + jD + \Delta_q$ with $\Delta_q \le D - 1$; $\Delta = \max_q \Delta_q = 17$; $S = \max_q s_q - \min_q s_q = 5$. Faults are attached to layers.

(1) Settling. If all faults of a set $F_<$ lie at layers $\le a$, the data frame they produce is constant from layer $a + D - 1$ on: only an ancilla fault can later change the data, through the remaining CNOTs of its own instance, all at layers $\le a + D - 2$; a freshly reset ancilla never pushes errors onto the data (an X-check ancilla only picks up Z errors from the data, a Z-check ancilla only X errors). Every instance reset after $a + D - 1$ is fault-free and measures $H_c e$. If all faults of $F_>$ lie at layers $\ge b$, measurements before b are unaffected.

(2) Splitting. Let $F = F_< \cup F_>$ be a weight-11 undetected logical with no fault strictly between a and b. A comparison detector (c, j) responds to $F_<$ only if $s(c, j-1) < a + D - 1$ and to $F_>$ only if the measurement of instance j, at $s(c, j-1) + D + \Delta_c \le a + 2D + \Delta - 2$, is at a layer $\ge b$. A first-round detector $(c, 0)$ responds to $F_>$ only if $b \le s_c + \Delta_c \le 21$, so with $a \ge 0$ it responds to both parts only if $b - a \le 21 < G$. A final detector responds to both only if $b - a < D + S + \Delta$. Hence for $b - a \ge G = \max(2D + \Delta - 1,\ D + S + \Delta) = 52$ no detector responds to both parts; each part is undetected, their logical effects add to a nonzero value, and one part has weight ≤ 10, contradicting Lemma 1. Consecutive faults of F are therefore less than G layers apart, and F spans at most $10(G-1) = 510$ layers.

(3) Deletion. The noisy SD6 circuit with R rounds equals the circuit with $R - 1$ rounds with an 18-layer block inserted at layer x, for every $x \in [4, 18(R-1)]$ (checked layer by layer in both bases; the template is exactly periodic). Removing a block $[x, x + D)$ that contains no fault of F removes one measurement of every ancilla; a detector of the shorter circuit that does not straddle the cut equals a detector of the longer one, and one that straddles it

equals the sum of the two detectors around the removed measurement. F therefore stays undetected, with the same logical effect, provided the frame entering layer $x + D$ of the longer circuit equals the frame entering layer x of the shorter one; this holds if $x + D \le f_1$ (all zero; f_1 the first fault layer) or $x \ge f_{11} + 2D - 2$ (data frame settled by (1), all ancillas in flight fault-free). *(4) Counting.* Write $f_1 \le f_{11}$ for the first and last fault layers of F; by (2), $f_{11} - f_1 \le 10(G - 1) = 510$. *Front:* while $f_1 \ge 4 + D = 22$, the block $[4, 22)$ is fault-free and can be deleted at $x = 4$, shifting F by -18 layers; afterwards $f_1 \le 21$ and hence $f_{11} \le 21 + 510 = 531$. *Back:* a block can be deleted at $x = f_{11} + 2D - 2$ as long as this x is an admissible insertion point, $x \le 18(R - 1)$. When no further deletion is possible, $18(R - 1) < f_{11} + 2D - 2 \le 531 + 34 = 565$, so $R - 1 \le 31$ and $R \le 32$. (Using $2D - 1$ in place of $2D - 2$, as a more conservative variant does, gives the same bound; the threshold $2D - 3$ would already suffice.) Thus every weight-11 undetected logical of a circuit with R rounds yields one of a circuit with at most 32 rounds, and with the lift computation of S13 none exists for any R.

For circuits that alternate with their time reverse, Shaw and Terhal reduce any number of rounds to a single round [27]; the gap argument above needs no such symmetry.

Circuits B and C. For circuit C, rounds are separated in time; a fault-free round inside the span of F splits it as in (2), so F occupies at most 11 consecutive rounds, and periodicity reduces any R to $R \le 13$. For circuit B we use a slab version of (2): a fault-free slab from the earliest reset to the latest measurement of one round splits F, provided that on every data qubit every propagating CNOT of round k inside the overlap with round $k + 1$ precedes every reading CNOT of round $k+1$. We verified this condition for all rounds of the circuits with $R = 3, 8, 24$ in both bases; it reduces any R to $R \le 23$.

S15. UPPER BOUND

For each basis we found a weight-12 logical operator of the gross code of the corresponding type with CP-SAT and applied it as single-qubit flips immediately after data initialisation; each flip is one SD6 mechanism. For $R = 1, 3, 6, 10$ and both bases (and, for circuit A, also at $R = 2$) no detector fires and the logical observable flips; omitting any one of the twelve flips makes the error detectable. Hence $d_{\mathrm{circ}} \le 12$ for circuits A, B and C.

S16. SEARCH AND OPTIMALITY

Search. The circuits were found with SAT models over schedules symmetric under the 24 colour-preserving translations. Variables: for each of the six classes, one of the eight orders of the Fig. 12 class, a start and six CNOT times; constraints: legality (six distinct times modulo P per data qubit), determinism (on every overlapping pair of X and Z checks, the number of shared qubits on which the X instance acts before the Z instance is even, for every instance offset), and, added lazily, one clause per frame-model-liftable weight-11 extended logical forbidding the timing that makes it liftable. Models were then certified with S2–S14. With the Fig. 12 orders pinned the model is unsatisfiable, in line with S9. As a control, 233 random legal deterministic schedules of the common-frame family, sampled without the blocking clauses, all have liftable weight-11 logicals.

Minimum period. With the same model and no circuit fixed, pipelined schedules with $P = 8, \ldots, 17$ and common-frame schedules with $P = 20$ are unsatisfiable. Every unsatisfiable clause set in the data set (129 in total: 104 for the obstruction and optimality instances with all logicals, and 25 length-2-only instances for Table S5) is accompanied by a DRAT proof produced by CaDiCaL and verified with drat-trim; the LRAT core was additionally verified with cake_lpr (SM S18). For the pipelined case a second, independent encoding (absolute times, one-sided difference literals) agrees for $P = 8, \ldots, 17$. The final clause sets were re-solved from scratch by independent SAT solvers (three for $P = 20$ common frame and $P = 12$, two for $P = 17$). Completeness of the family encoding: at $P = 8$ it enumerates exactly the 944 legal deterministic circuits found by an independent backtracking search, and 75 independently sampled legal deterministic circuits at $P = 12, 17$ (pipelined) and 20 (common frame) are all models. Soundness of the clauses: in 991 sampled cases, including logicals with several residual columns, every lift asserted by the frame model was confirmed on the exact DEM, and in 99 sampled cases every violated clause corresponded to an exact lift. Unsatisfiability means that every legal deterministic circuit of the family has a model-liftable weight-11 logical. For logicals of ten data errors and one length-2 residual this implies a genuine weight-11 undetected error for sufficiently many rounds (S9); for logicals with several residual columns the same implication is confirmed in the sampled cases above but not proven in general. The minimum periods are therefore established within the model and refer to sufficiently many rounds. Restricting the CNOT window of class X_0 or of class Z_2 to at most 14 steps is unsatisfiable at $P = 24$, while each of the other four classes can be so restricted; since a window spans at most $P - 3$ steps, this suggests why $P \ge 18$ is

needed, but the bound itself rests on the direct runs. At $P = 8$ all 944 legal deterministic circuits of the family were also checked directly: each has a weight-11 undetected error at $R = 1$ in the Z basis.

Uniqueness of the order class. Per check, the extended-code columns are the data columns and three residual sets; up to reversal there are 90 types per basis, of which 45 contain a residual of length 3 that, together with nine data columns, forms a weight-10 extended logical. For each of the 24 colourings by cosets of translation subgroups of index 2, 3, 4, 6, 8 and 9 (3, 4, 3, 12, 1 and 1 subgroups), with one type per colour, a SAT search with witness clauses leaves only the translates of Fig. 12 with $d_{\text{ext}} \geq 11$ in both bases; every witness used was re-verified as at most ten distinct columns of that assignment's extended code with vanishing H part and nonzero L part. Colourings of index 12, 18, 24 and 36 and per-check-free assignments are not covered.

Less symmetry. In the same model, pipelined schedules invariant under the order-12 subgroup generated by $(1, 2)$ are obstructed for every $P \leq 11$, and schedules invariant under any subgroup of order ≥ 2 are obstructed at $P = 8$. Schedules without symmetry, and the remaining subgroups at $P = 12, \ldots, 17$, are open.

S17. SENSITIVITY TO IDLE NOISE

Noise model. Starting from SD6 we multiply only the idle depolarizing channel by λ, $p_{\text{idle}} = \lambda p$; $\lambda = 1$ reproduces SD6 exactly and $\lambda = 0$ coincides with circuit-level depolarizing noise without idle errors (both checked on the detector error models). The CX-native variant uses SI1000 coefficients with CX, RX and MX treated as native operations; it is not the standard SI1000 model. Compiling CX to $H\,\mathrm{CZ}\,H$ and using Z-basis resets and measurements would raise the per-round fault mass of the period-18 circuit by about 12% and that of the period-8 circuits by 6–9%, so the variant slightly favours the period-18 circuit.

Direct sampling. $R = 12$, identical BP-OSD and Relay-BP decoders with exact priors, both bases. Table S9 lists per-round ratios of the logical error rate of the period-18 circuit to Fig. 12 and to the IBM circuit, with 95% intervals. At $\lambda = 0$ and $\lambda = 0.1$, $p = 4 \times 10^{-3}$, an independent re-sampling with a different noise implementation and BP-OSD code gave 0.37 [0.28, 0.49] and 2.73 [2.14, 3.47] for the ratio to Fig. 12. Several entries rest on few failures; at $p = 2 \times 10^{-3}$ and $\lambda = 0$ no circuit can be distinguished.

TABLE S9. Per-round logical error rate of the period-18 circuit divided by that of Fig. 12 and of the IBM circuit (95% intervals; "–": too few failures).

noise	p	to Fig. 12, BP-OSD	to Fig. 12, Relay-BP	to IBM, BP-OSD	to IBM, Relay-BP
$\lambda = 0$	4×10^{-3}	0.34 [0.19, 0.59]	0.25 [0.11, 0.60]	0.89 [0.45, 1.7]	1.0 [0.33, 2.9]
$\lambda = 0$	3×10^{-3}	0.40 [0.16, 1.02]	0.40 [0.09, 1.9]	1.0 [0.34, 3.0]	0.50 [0.11, 2.4]
$\lambda = 0.1$	4×10^{-3}	3.0 [1.9, 4.7]	2.4 [1.3, 4.5]	3.7 [2.2, 6.0]	7.8 [2.9, 20]
$\lambda = 0.1$	3×10^{-3}	2.5 [0.81, 7.5]	4 [0.56, 22]	3.3 [0.96, 11]	–
$\lambda = 0.1$	2×10^{-3}	2.3 [1.09, 4.95]	–	1.1 [0.19, 4.9]	–
$\lambda = 0.3$	4×10^{-3}	33 [22, 50]	73 [42, 124]	44 [28, 68]	164 [79, 339]
$\lambda = 0.3$	3×10^{-3}	15 [6.9, 33]	20 [7.6, 51]	13 [6.3, 27]	–
CX-native	2×10^{-3}	23 [13, 39]	40 [22, 74]	36 [19, 70]	44 [23, 83]

Full SD6 noise ($\lambda = 1$). Table S10 lists the per-round logical error rates (both bases summed, 95% intervals) with the numbers of failures. At $p = 2 \times 10^{-3}$ an independent re-sampling with 4×10^5 shots for the IBM circuit (22 and 25 failures in the two bases) and 400 per basis for the period-18 circuit (66 and 67 failures) gives the ratio 1.52×10^3 [1.10×10^3, 2.14×10^3]. At $p = 10^{-3}$ neither period-8 circuit failed in 20000 shots, so the ratio of at least 3 is the period-18 rate divided by the 95% upper limit of the IBM rate: a lower bound, not an estimate. The factor of about 1.5×10^3 at $p = 2 \times 10^{-3}$ and the lower bound of 3 at $p = 10^{-3}$ refer to different p and are consistent.

Per-layer rates. Dividing per-round rates by the period treats every layer as having unit duration, consistent with SD6 but not with physical time. Each round of the period-18 circuit has 12 CX-only layers and 6 layers with measurements or resets; Fig. 12 has 2 and 6. If measurement and reset layers are much longer than CX layers, the ratio of round durations falls from 2.25 towards 1.1–1.3, and the per-layer advantage of the longer period largely disappears.

Leading order. For an ideal decoder at low p, the per-round rate is dominated by weight-12 logical errors with total weight $A_6 p^6$. Summing over the enumerated families (weight-11 logicals and one type of weight-12 logicals), the ratio $A_6(\text{period } 18)/A_6(\text{Fig. } 12)$ is 0.0104 at $\lambda = 0$, 0.084 at $\lambda = 0.1$, 0.84 at $\lambda = 0.3$ and 26 at $\lambda = 1$, crossing 1 near

TABLE S10. Per-round logical error rates under full SD6 noise ($\lambda = 1$), $R = 12$, both bases summed, with failure counts (bases Z+X) where available; "<": 95% upper limit with no failure.

p	decoder	period 18	Fig. 12	IBM
10^{-3}	BP-OSD	9.1×10^{-5} [6.2, 13.4]$\times 10^{-5}$ (15+10)	0/20000, $< 3.1 \times 10^{-5}$	0/20000, $< 3.0 \times 10^{-5}$
10^{-3}	Relay-BP	1.1×10^{-5} [0.40, 3.1]$\times 10^{-5}$ (0+3)	0/20000, $< 3.1 \times 10^{-5}$	0/20000, $< 3.1 \times 10^{-5}$
2×10^{-3}	BP-OSD	3.08×10^{-2} [2.69, 3.54]$\times 10^{-2}$ (100+100)	3.3×10^{-5} [1.4, 8.4]$\times 10^{-5}$ (1+3)	2.1×10^{-5} [0.8, 6.3]$\times 10^{-5}$ (1+2)
2×10^{-3}	Relay-BP	1.44×10^{-2} [1.26, 1.65]$\times 10^{-2}$	0/20893, $< 3.0 \times 10^{-5}$	7.3×10^{-6} [0.18, 4.1]$\times 10^{-5}$ (1+0)

$\lambda \approx 0.32$. Caveats: A_6 is a union-type estimate, not a bound; the other type of weight-12 logical is not enumerated for either circuit, and an unknown factor F in their relative weight moves the crossing (to 0.50 for $F = 0.3$, 0.24 for $F = 2$, 0.20 for $F = 3$); the estimate describes $p \lesssim 10^{-4}$, not the sampled range (at $\lambda = 0.3$ it predicts near equality while direct sampling at $3\text{–}4 \times 10^{-3}$ shows a factor 15–73); and it says nothing about practical decoders.

S18. VERIFICATION AND DATA

All circuits were built and simulated with Stim [18]; exact feasibility questions were decided with CP-SAT and with CDCL SAT solvers, and the extended-code searches use a dedicated depth-first search. The following were obtained with two independent implementations that agree: the construction of the extended code and $d_{\text{ext}} = 11$; the list of 4224 weight-11 logicals; the lift computations; the enumeration of period-8 schedules; and the minimum-period runs. The arguments of Lemma 1 and of the round reductions were checked independently, and their structural premises (template identity, image set, periodicity, detector structure) were verified by computer for every circuit. The data set (Data availability) contains the circuit specifications (circuits A, B, C and Fig. 12), the lists of 4224 weight-11 extended logicals for each basis, the witnesses (the weight-12 upper-bound witnesses and the weight-11 witnesses for Fig. 12 and its retimings), the certificates (clause sets of the unsatisfiable instances), and the reproduction scripts for the lower-bound computations; it is available at [Zenodo DOI]. Every unsatisfiable clause set in the data set (129 in total: 104 for the obstruction and optimality instances with all logicals, and 25 length-2-only instances for Table S5), and every batch of the lift computation, is accompanied by a DRAT proof produced by CaDiCaL and verified with drat-trim; the LRAT core was additionally verified with cake_lpr. Here "accompanied" means: proofs of up to 100 MB are included in the data set; larger ones are identified by their SHA-256 hash and can be regenerated byte-identically with the included script. The certificates cover the solver decisions only; the circuit template and the reduction of Lemma 1 are covered by the pen-and-paper proofs and code-level checks of S2. Limitation: $d_{\text{ext}} = 11$ and the completeness of the list of 4224 weight-11 logicals rest on two independent exhaustive searches; we did not obtain a checkable proof certificate for them (a direct CNF encoding is beyond CDCL solvers already at weight 6).

S19. THE MORPHING END-CYCLE CODE IS NOT THE GROSS CODE

The end-cycle code of the Shaw–Terhal morphing circuit derived from the $[[288, 12, 18]]$ code with homomorphism f_x [9] has parameters $[[144, 12, 12]]$: $k = 12$, and an exhaustive search finds no logical of weight ≤ 11 and one of weight 12 for both error types. Its stabiliser group has minimum weight exactly 9: an exhaustive search over the nonzero elements of the row spaces of H_X and H_Z finds none of weight ≤ 8 and one of weight 9, for both types. The gross code has stabiliser generators of weight 6. The minimum weight of a CSS stabiliser group is the smaller of the minimum X- and Z-type weights, and it is invariant under qubit permutations and local Clifford operations, which map stabilisers to stabilisers of the same weight. The two codes are therefore not equivalent under these operations; the argument says nothing about more general equivalences.